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(SDIA)

EnergyAI Multi-Horizon Electricity Consumption Forecasting

Methodological Audit, Deep Learning Benchmarking, and
Operationalization in a Reliable Platform

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Abstract

Electricity-consumption forecasting is useful only when reported performance reflects genuinely unseen future conditions. This project began with the replication of a published Deep Energy Predictor Model combining a cascaded residual network, a deep neural network, and XGBoost. Although the initial implementation reproduced classification accuracy close to the values announced in the article, a methodological audit showed that the task and validation protocol allowed target information to enter the predictors. The most important mechanisms were direct use of a target-derived variable, the physical relationship between active power, voltage, and current, non-shifted rolling features, preprocessing before data separation, and random splitting of strongly correlated temporal observations. The historical high scores were therefore retained as diagnostic evidence but excluded from forecasting claims.

The problem was reformulated as continuous multi-horizon forecasting with causal windows, chronological train-validation-test separation, train-only fitting of preprocessing operations, inverse transformation, physical-unit metrics, and seasonal baselines. The research then evaluated recurrent, convolutional, patch-based, inverted-Transformer, global, and pretrained foundation-model approaches across IHEPC, Steel Industry, ECL, Low Carbon London, Tetouan, and external diagnostic data. The deployed Global Temporal Fusion Transformers retain strong 24-hour and 168-hour London cold-start evidence. A new 30-day Chronos-2 model, adapted through rank-8 LoRA and conformal calibration, improves the strongest frozen Tetouan selection baseline by 49.82%, reaches macro $R^2 = 0.4498$, and obtains 88.15% central-80% interval coverage. Separate Chronos-2 hourly challengers improve on the packaged TFTs by 2.90% and 2.62% on the frozen London cohort, but remain promotion-ready rather than deployed because their modest gains do not yet justify the added memory and input-history contract.

The repository release integrates the two hourly TFTs and the 30-day Chronos-2 adapter into EnergyAI, a Dockerized Next.js and FastAPI platform with PostgreSQL persistence, authenticated ownership, bounded CSV ingestion, explicit hourly-versus-daily cadence metadata, forecast history, analytics export, and readiness checks. The production image installs the pinned foundation-model runtime, verifies artifact and base-model checksums, and caches the exact Chronos-2 base for offline startup. The verified public Azure release now serves the 24-hour, 168-hour, and 30-day artifacts; its readiness payload reports all three available, enabled, and warmed. Durable avatar storage, cloud alert and avatar-cleanup workers, centralized logging, restore testing, and Moroccan-site efficacy remain outside the verified scope.

Keywords: electricity consumption forecasting; data leakage; time series; Temporal Fusion Transformer; cold-start evaluation; reproducibility; EnergyAI.

Résumé

La prévision de la consommation électrique n'est utile que lorsque les performances annoncées reflètent réellement des conditions futures non observées. Ce projet a débuté par la réplique d'un Deep Energy Predictor Model combinant un réseau résiduel en cascade, un réseau neuronal profond et XGBoost. Bien que la première implémentation ait reproduit une exactitude de classification proche des valeurs annoncées dans l'article, un audit méthodologique a montré que la tâche et le protocole de validation permettaient à des informations dérivées de la cible d'entrer dans les variables explicatives. Les principaux mécanismes identifiés sont l'utilisation directe d'une variable liée à la cible, la relation physique entre puissance active, tension et intensité, des fenêtres glissantes non décalées, le prétraitement avant séparation et une division aléatoire d'observations temporelles fortement corrélées. Les scores historiques élevés sont donc conservés comme éléments de diagnostic, mais exclus des contributions de prévision.

Le problème a été reformulé en prévision continue multi-horizon avec fenêtres causales, séparation chronologique entraînement-validation-test, ajustement des prétraitements sur l'entraînement uniquement, transformation inverse, métriques en unités physiques et baselines saisonnières. L'étude compare aussi des modèles de fondation préentraînés. Le modèle mensuel Chronos-2, adapté par LoRA de rang 8 et calibré par une méthode conforme, améliore de 49,82% la meilleure baseline figée de sélection sur Tétouan, atteint un R^2 macro de 0,4498 et une couverture de 88,15% pour l'intervalle central nominal de 80%. Des challengers Chronos-2 horaires améliorent les TFT de 2,90% et 2,62% sur la cohorte londonienne figée, mais restent non déployés car le gain ne justifie pas encore le coût mémoire et un historique d'entrée plus long.

La version actuelle du dépôt intègre les deux TFT horaires et l'adaptateur Chronos-2 à 30 jours dans EnergyAI. Le contrat distingue explicitement 24 et 168 cibles horaires de 30 cibles quotidiennes. L'image de production vérifie les sommes de contrôle et met en cache la version exacte du modèle de base pour un démarrage hors ligne. La version publique Azure vérifiée sert désormais les artefacts de 24 heures, 168 heures et 30 jours ; l'état de préparation confirme que les trois sont disponibles, activés et chargés. Le stockage durable des avatars, les workers cloud d'alertes et de nettoyage, la journalisation centralisée, les tests de restauration et l'efficacité sur des sites marocains restent hors du périmètre vérifié.

Mots-clés : prévision de consommation électrique ; fuite de données ; séries temporelles ; Temporal Fusion Transformer ; Chronos-2 ; LoRA ; reproductibilité ; EnergyAI.

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List of Acronyms

Acronym	Meaning
API	Application Programming Interface
AUC	Area Under the Receiver Operating Characteristic Curve
CNN	Convolutional Neural Network
DEPM	Deep Energy Predictor Model
DNN	Deep Neural Network
ECL	ElectricityLoadDiagrams20112014
EECP	Electric Energy Consumption Prediction
FFT	Fast Fourier Transform
GRU	Gated Recurrent Unit
IHEPC	Individual Household Electric Power Consumption
JWT	JSON Web Token
LCL	Low Carbon London
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MASE	Mean Absolute Scaled Error
ML	Machine Learning
PCA	Principal Component Analysis
RBAC	Role-Based Access Control
RevIN	Reversible Instance Normalization
RMSE	Root Mean Squared Error
RNN	Recurrent Neural Network
sMAPE	Symmetric Mean Absolute Percentage Error
TFT	Temporal Fusion Transformer
UML	Unified Modeling Language
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

General Introduction

Electricity must be generated, transported, and consumed in near real time. Forecast errors therefore have practical consequences: they can distort purchasing decisions, reserve planning, demand-response actions, and household budget estimates. At the same time, electricity demand is difficult to predict because it combines several temporal scales. Daily routines, weekends, holidays, weather, appliance usage, occupancy, and structural changes interact with one another. A useful forecasting system must consequently do more than fit a historical curve. It must state what is predicted, how far into the future, from which information, and with what evidence that the reported error will generalize.

This project began from an apparently straightforward objective: reproduce the Deep Energy Predictor Model (DEPM) proposed by Ragupathi et al. [1]. The article describes a hybrid architecture that combines XGBoost, a Deep Neural Network (DNN), and a cascaded residual network and reports 98% classification accuracy on household electricity data. The first local experiments reconstructed the pipeline and obtained similarly high scores. If read without further investigation, this outcome might have been presented as a successful replication or even an improvement. However, an elementary logistic-regression diagnostic produced 99.4% accuracy when variables capable of reconstructing the target were included. That result redirected the project toward a more important question: did the models learn future demand, or did the evaluation reveal the answer to them?

The audit identified several distinct failure mechanisms. In early local implementations, the class label was derived from active power while active power itself, or mathematically related variables, remained among the predictors. Voltage and global intensity also encode an approximate reconstruction of power through the physical relationship $P \approx V \times I$. Rolling statistics sometimes included the current target value. Standardization and Principal Component Analysis (PCA) were fitted before the split, allowing test-distribution information into the training pipeline. Finally, random splitting placed temporally adjacent and highly correlated measurements on both sides of the evaluation boundary. The resulting score measured neither a genuine future forecast nor deployment-time generalization.

The project was therefore reformulated around continuous, causal, multi-horizon forecasting. Inputs were restricted to information available at prediction time; windows were shifted; train, validation, and test regions were ordered chronologically; preprocessing was fitted on training observations; and metrics were evaluated after transformation back to physical units. Seasonal-naïve and persistence forecasts were treated as mandatory baselines. Model selection used validation evidence, while the test cohort was reserved for final comparison. This methodological change is the central contribution of the work,

because it turns an initially misleading high score into an auditable forecasting problem.

The corrected research path evaluated several model families. Recurrent and convolutional approaches included Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), bidirectional variants, and CNN-BiLSTM, drawing in particular on the EECF-CBL architecture of Le et al. [2]. Transformer-oriented experiments included PatchTST [3], iTransformer [4], xPatch [5], and the pretrained Chronos family [6], [7]. More recent TimePro and TimeMixer++ integrations were investigated, but the report does not claim successful reproduction where no completed metric artifact exists. Product selection combines Global TFT and N-BEATS household experiments with a frozen Chronos-2 plus LoRA protocol.

The repository release exposes three explicit prediction contracts: 24 and 168 hourly values from the compact Global TFTs, and 30 daily values from a Chronos-2 120M model adapted through rank-8 LoRA. The TFT results retain their original London cold-start evidence. The new month model was selected without opening its transfer diagnostics, then audited on Portugal and southern Morocco. It improves the strongest frozen Tetouan selection baseline by 49.82%, while the positive Portugal result and negative southern-Morocco result make the geographic boundary visible rather than hiding it. Applying the same architecture to the hourly tasks produced statistically broad but operationally modest improvements over TFT, so those adapters are packaged as production-eligible challengers and are not active in the application.

The selected models were integrated into EnergyAI, a full-stack platform that combines a Next.js frontend, a FastAPI backend, PostgreSQL persistence, background workers, authenticated data ownership, CSV ingestion, forecast history, alerts, recommendations, analytics-PDF export, and model-readiness checks. Each packaged artifact has a manifest, a SHA-256 checksum, an input contract, data-quality gates, an output shape, and documented limitations. The audit also identified an important evidence boundary: the frozen research evaluation uses per-household normalization fitted on each household's training segment, whereas serving uses statistics from the latest accepted 336-hour window. The software path exists and is testable, but the frozen headline metrics do not validate that changed serving transform. This distinction is maintained throughout the report.

Research questions

The PFE is organized around four questions:

- RQ1** Which target, feature, preprocessing, and split choices made the initial DEPM classification scores unsuitable as evidence of future electricity-consumption forecasting?

- RQ2** Under causal chronological evaluation, which completed deep-model families provide the strongest evidence for the benchmark task and for future-time transfer to households excluded from gradient updates?
- RQ3** How can selected forecasting checkpoints be operationalized in a user-facing platform with explicit data, artifact, ownership, readiness, and uncertainty contracts?
- RQ4** Which scientific and operational claims remain unsupported, especially across geography, random seeds, uncertainty calibration, and research-to-serving preprocessing?

Demonstrated contributions

The resulting contributions are:

1. a forensic reconstruction of the supervisor-supplied starting paper and early local implementations, preserving high scores as historical evidence while identifying direct, proxy, preprocessing, window, and split leakage;
2. a corrected causal protocol for continuous multi-horizon forecasting, including physical-unit metrics, seasonal baselines, shared chronology, and known-versus-cold-start household evaluation;
3. a traceable experimental survey spanning recurrent, convolutional, patch-based, inverted, decomposition, and global approaches, with failed or incomplete integrations reported separately from completed results;
4. checkpoint-backed Global TFT analysis at 24 and 168 hours and a frozen Chronos-2 plus LoRA month-selection protocol with independent transfer diagnostics;
5. a serious same-architecture hourly challenger study whose models pass the frozen accuracy gates but remain undeployed under an explicit resource-and-contract decision;
6. an implemented EnergyAI platform whose principal routes, API operations, access controls, ingestion workflow, artifact checks, and current validation limits are mapped to repository evidence; and
7. an explicit research–serve normalization distinction, evaluated on the exact preserved cold-start cohort and retained as a transfer limitation beyond that cohort.

The research trajectory is summarized in figure 1. The report is organized into seven chapters. Chapter 1 defines the problem, requirements, stakeholders, and product boundary. Chapter 2 provides a reasoned review of forecasting model families and validation risks. Chapter 3 reconstructs the DEPM and presents the methodological audit. Chapter 4 defines the corrected experimental protocol. Chapter 5 reports evidence-backed experiments and final model selection. Chapter 6 documents the EnergyAI design and implementation. Chapter 7 consolidates validation results, limitations, and threats

to validity.

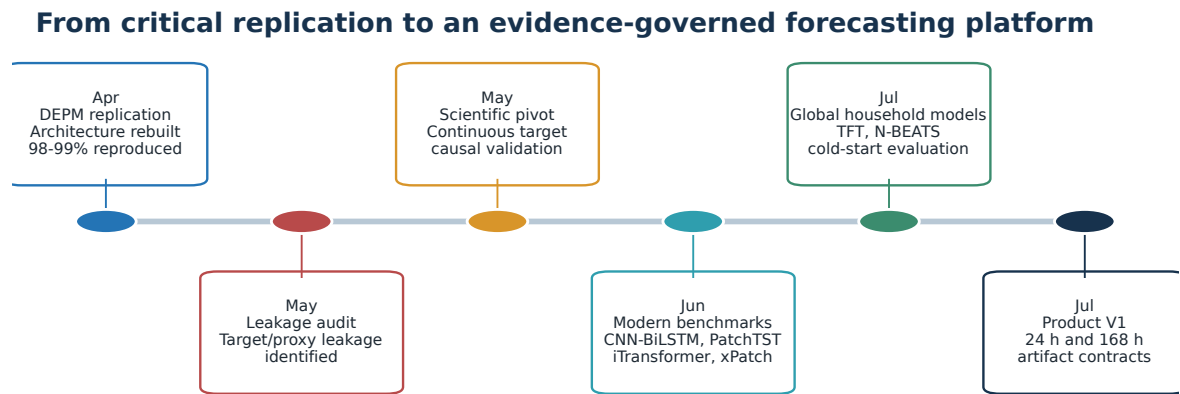


Figure 1 – Research trajectory from DEPM replication to the EnergyAI Product V1 checkpoint. Source: Git timeline frozen on 31 July 2026.

The main thesis of this report is deliberately conservative: a lower but causally valid error is more valuable than a spectacular score obtained through target reconstruction or an unrealistic split. EnergyAI is presented not as a claim of perfect prediction, but as a traceable system in which scientific conclusions, model artifacts, software behavior, and remaining limitations can all be inspected.

Context, Requirements, and Project Framing

1.1 Electricity consumption as a forecasting problem

Electricity consumption is a time-dependent process whose distribution changes with the hour of day, day of week, season, household behavior, tariff incentives, and external conditions. Forecasting differs from ordinary regression because the ordering of observations is part of the problem. A model evaluated on a random sample of the same historical period may perform well while failing on the next day. The operational question is therefore conditional and time-indexed: given the measurements available up to time t , estimate consumption for times $t + 1$ through $t + H$.

For a univariate household series y_t , a direct multi-horizon forecaster can be written as

$$(\hat{y}_{t+1}, \dots, \hat{y}_{t+H}) = f_{\theta}(y_{t-L+1:t}, c_{t-L+1:t+H}), \quad (1.1)$$

where L is the lookback length, H the forecast horizon, and c represents calendar or other covariates known at forecast time. The packaged hourly TFTs use 336 hours and output 24 or 168 hourly values. The month model integrates persisted intervals into at least 270 valid rolling daily totals, uses up to 365 daily values, and outputs 30 daily values. All active models expose the 0.1, 0.5, and 0.9 quantiles.

This formulation must not be confused with a binary label such as “high” or “low” consumption. Classification can support anomaly or state detection, but it does not answer how many kWh will be consumed at each future hour. It can also become trivial if the class is constructed from a measurement retained among the inputs. One of the first framing decisions of this PFE was therefore to replace the historical class objective with a physically interpretable continuous target.

1.2 Operational horizons and user value

EnergyAI exposes three supported capabilities. The 24-hour horizon is intended for next-day review, and the 168-hour horizon supports week-ahead planning; both preserve hourly structure. The 30-day capability emits 30 daily totals rather than pretending that an integer duration of 720 means 720 hourly targets. It supports month-ahead

budget planning and carries explicit daily cadence, target-count, input-readiness, and provenance metadata. Earlier monthly N-BEATS and MultiCycleNet results remain historical; the active month capability is a separately selected and audited Chronos-2 release.

The value of a forecast depends on its comparison target. Repeating yesterday’s load or the same hour from the previous week often yields a surprisingly strong baseline. A deep model is only justified when it consistently improves over such rules, not merely when it reports a low number on an arbitrary normalized scale. This report consequently compares Global TFT with a seasonal-naive baseline and reports both average error and the proportion of households for which the model wins.

1.3 Stakeholders

The principal stakeholders are summarized in table 1.1. The platform is designed around an authenticated user who owns a site and its meter data. The administrator manages accounts and inspects system and model readiness. The supervisor and academic jury are evidence reviewers rather than product users: they need to verify the scientific narrative, result provenance, and limitations. Future energy managers or household users would require additional validation on their own local data before operational reliance.

Table 1.1 – Stakeholder needs and associated evidence.

Stakeholder	Primary need	Evidence expected
Normal user	Import consumption, inspect usage, generate supported forecasts, review actions	Clear data validation, forecast provenance, uncertainty, history, ownership isolation
Administrator	Operate accounts and inspect platform/model readiness	Role checks, health endpoints, audit events, artifact status
Supervisor and jury	Assess scientific rigor and engineering contribution	Reproducible protocol, primary citations, manifests, tests, Git timeline
Future operator	Deploy and monitor a stable system	Configuration, backups, dependency policy, observability, deployment validation
Data subject	Retain control over personal consumption data	Authentication, export, deletion, minimal exposure, bounded ingestion

1.4 Functional scope

The current repository supports the following primary user flow:

- FR-01** create or access an account through the available authentication capability;
- FR-02** complete site, timezone, tariff, and meter setup;
- FR-03** preview and import a CSV containing timestamps and energy values;
- FR-04** inspect consumption summaries and paginated readings;
- FR-05** verify model readiness, choose 24 hours, 168 hours, or 30 daily targets, and generate a forecast with quantiles, cadence, and provenance;
- FR-06** review alerts and recommendation-based actions;
- FR-07** manage site, budget, account export, and account deletion settings; and
- FR-08** permit administrators to manage users and inspect system/model readiness without exposing that surface to normal users.

Routes and endpoints exist for several secondary or gated behaviors, including Google authentication, email verification, password recovery, simulation controls, and administrative monitoring. This report describes them only to the degree supported by current code and configuration. It does not present removed navigation items or hidden research surfaces as primary Product V1 features.

EnergyAI use-case diagram

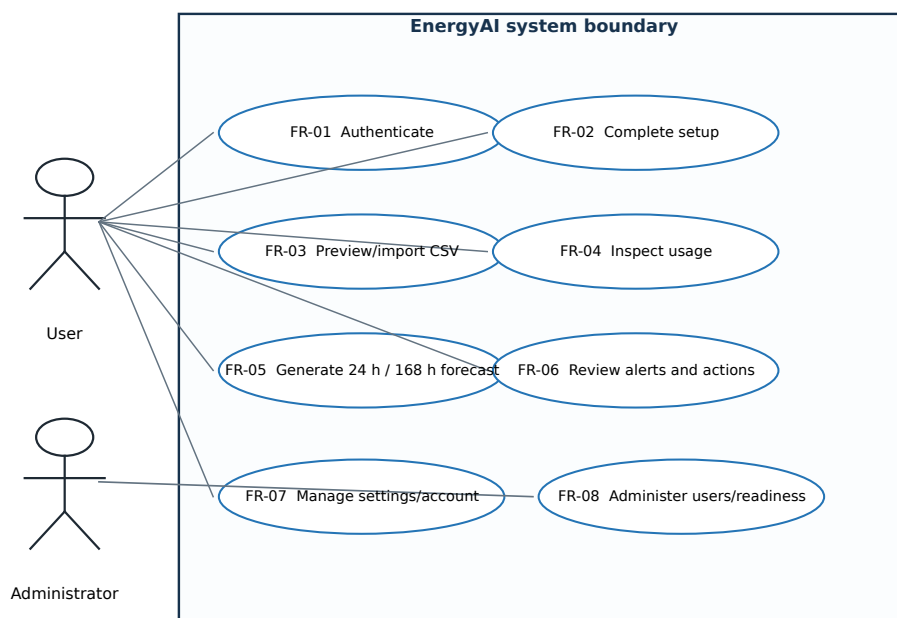


Figure 1.1 – EnergyAI use-case diagram derived from current frontend routes and backend routers.

1.5 Non-functional requirements

1.5.1 Reliability and truthfulness

The application must refuse unsupported operations rather than fabricate a result. Forecast readiness is a first-class contract. The backend checks artifact presence, checksum, input coverage, maximum consecutive gaps, finite values, and supported horizon before inference. Failure is returned as an explicit state that the frontend can explain.

1.5.2 Privacy and isolation

Electricity traces can reveal occupancy and routine. Each Site and Meter is associated with an owning user. API queries must apply ownership conditions, and administrative access must be explicit. Authentication uses short-lived access tokens and refresh credentials stored as hashes server-side; the browser receives refresh state through an HttpOnly cookie rather than a JSON token saved in web storage.

1.5.3 Latency and resource constraints

The user experience requires bounded imports and responsive inference. The product therefore packages fixed CPU-loadable artifacts instead of retraining in response to user requests. The two TFT checkpoints are approximately 5.6 MB each. The month LoRA adapter is approximately 4.9 MB, but its pinned 120M-parameter base adds a substantial image-cache and memory cost. A measured CPU cold load took about 9.0 seconds; warm 30-day inference took about 77 ms. The production image therefore caches the exact base revision and verifies both base and adapter checksums before offline serving.

1.5.4 Cold-start generalization

Known-household performance can hide dependence on household identity. A product that accepts a new user’s data must also be assessed on households that contributed no gradient updates. The Low Carbon London experiments split entire households into known and cold-start cohorts before model training. Checkpoint-selection claims in this report prioritize cold-start macro metrics, even when a global metric appears higher.

1.5.5 Reproducibility

Every central model claim must map to a manifest or leaderboard, and every packaged checkpoint must map to a SHA-256 digest. The report itself maintains registries for models, figures, notebooks, commits, artifact hashes, tests, and requirement-to-test

traceability. This requirement is stricter than merely retaining source code: it connects the exact file and experimental status to each published claim.

1.6 System boundary and assumptions

EnergyAI retains a reproducible six-service Docker Compose configuration for local development and controlled demonstrations. The jury release was updated through Git commit `6796b1ec7360f9c05cb804fbd8be29ba1ea78ef8` and deployed to Microsoft Azure on 9 August 2026. The verified cloud boundary comprises public Next.js and FastAPI Container Apps, a dedicated email-worker Container App, Azure Database for PostgreSQL, Azure Container Registry, and Azure Key Vault. Azure-managed hostnames provide HTTPS for the web and API endpoints. Public email registration and verification, SMTP delivery, and Google identity are enabled and smoke-tested. The deployed worker scope remains narrower than local Compose: alert and avatar-cleanup workers are absent, and avatar storage is ephemeral. Likewise, validation metrics describe London and benchmark cohorts; public availability does not guarantee performance on a Moroccan household or organization without a local evaluation.

The cover page uses the official combined UMI/FSM logo and blue-orange motif published by the Faculty of Sciences of Meknes. The department and programme wording were verified against the current institutional programme pages, and the faculty identifies the academic supervisor as Pr. Ali Oubelkacem. No official defense-date evidence is preserved in the project, so the date remains an explicit administrative placeholder. The publicly available cover model is intended for doctoral theses rather than Master PFE reports; consequently, the English PFE wording is an evidence-based adaptation, while the jury composition also remains pending confirmation.

1.7 Project risks and acceptance criteria

Table 1.2 – Principal risks and acceptance criteria.

Risk		Control	Acceptance evidence
Target leakage		Causal feature audit and chronological split	Reproducible comparison between historical and corrected pipelines
Misleading model comparison		Common cohorts, units, horizons, and baselines	Leaderboards with explicit protocol columns
Unsupported product promise		Capability, cadence, and readiness gates	24 and 168 hourly targets plus exactly 30 daily targets
Cross-user data exposure		Ownership-scoped queries and RBAC	PostgreSQL integration and authorization tests
Corrupted or mismatched model		Manifest and SHA-256 verification	Strict load, finite output, and expected shape checks
Research-serving transform mismatch		Explicit normalization contracts and frozen-cohort serving re-evaluation	Transform comparison completed; new-site efficacy remains conditional
Dependency or browser gate failure		Automated quality and security audits	Current QA report including conditional findings
Institutional non-compliance		Official template and administrative review	Placeholder closure before final submission

This framing makes the scientific and product objectives mutually reinforcing. Reliable software cannot repair an invalid experiment, while a valid model cannot create user value without data controls, understandable output, and operational evidence.

A Reasoned Review of Forecasting Approaches

2.1 From prediction accuracy to forecasting validity

The literature on energy forecasting contains models with very different targets, horizons, sampling rates, and evaluation scales. A numerical score cannot be interpreted without this context. An accuracy percentage may describe a high/low class, while Mean Absolute Error (MAE) may be expressed in normalized units, kW, kWh, or an aggregate over clients. A model can predict one point recursively or produce an entire horizon directly. These distinctions determine whether two results are comparable.

The review in this chapter is therefore organized by the forecasting difficulty addressed by each family, rather than as independent summaries of papers. It also treats validation design as part of the model: a sophisticated architecture cannot compensate for a split that exposes future information.

2.2 Statistical and seasonal baselines

Persistence forecasts set $\hat{y}_{t+h} = y_t$ for all h or reuse a recent observation appropriate to the sampling frequency. Seasonal-naive forecasts use the last corresponding seasonal value, such as $\hat{y}_{t+h} = y_{t+h-s}$ with daily or weekly period s . These methods have no learned representation, but they exploit the strongest regularity in many electricity series. They are inexpensive, transparent, and difficult to beat consistently for stable households.

Classical exponential smoothing and autoregressive models provide trend, seasonality, and residual dynamics with explicit assumptions. Their limitations arise when many nonlinear interactions, large multivariate panels, or long direct horizons are involved. Nevertheless, they remain essential references because a model whose absolute error is low but worse than seasonal naive has not demonstrated practical added value [8].

2.3 Classical machine learning

Linear and regularized regressions can combine lagged values, calendar terms, and rolling statistics. Tree ensembles such as random forests and XGBoost capture nonlinear thresholds and interactions and often perform well on engineered tabular features; XGBoost’s original system combines regularized boosting with sparsity-aware and scalable learning mechanisms [9]. Their principal risk in time-series work is not architectural but procedural: current or future target information can be hidden in aggregates, centered windows, or derived physical variables.

In this project, logistic regression became a diagnostic instrument. When it achieved 99.4% classification accuracy with the suspect variables, it demonstrated that high DEPM accuracy did not require the hybrid neural architecture. This is a general lesson: a deliberately simple baseline can expose an ill-posed task more effectively than another deep model.

2.4 Recurrent sequence models

A recurrent neural network (RNN) maintains a hidden state that summarizes prior inputs, but simple recurrent units can suffer from vanishing or exploding gradients. LSTM and GRU architectures introduce gates that regulate memory updates and improve learning over longer dependencies [10], [11]. Bidirectional variants process a training sequence in both directions. They are valuable for representation learning when the complete input window is known, but their use must remain causal with respect to the prediction target: a backward recurrence over the historical input window is acceptable, while access to future target samples is not.

The local research archive contains standalone DNN, LSTM, BiLSTM, GRU, and BiGRU classifiers, as well as DEPM hybrids using their representations with XGBoost. After removal of direct target information, the recorded classification accuracies cluster approximately between 0.885 and 0.896 rather than around 0.99. This comparison supports the audit, but it remains a classification study and is not used as the final forecasting leaderboard.

2.5 Convolutional-recurrent hybrids

One-dimensional convolutions can extract local motifs and reduce noise before recurrent processing. Le et al. propose EECP-CBL, a CNN-BiLSTM architecture evaluated on minutely, hourly, daily, and weekly forms of IHEPC [2]. The design is attractive when both local fluctuations and longer temporal context matter. The paper reports

comparisons with linear regression, LSTM, and CNN-LSTM using MSE, RMSE, MAE, MAPE, training time, and prediction time.

The PFE used this work as the first structured pivot away from DEPM classification. Local notebooks implemented chronological splitting, train-only scaling, continuous targets, inverse transformation, and non-shuffled evaluation. The exact published numbers are not claimed as reproduced unless matched by a preserved local output. The CNN-BiLSTM family also appears as a baseline in the later ECL benchmark, where it is evaluated under the same split and scaling policy as the Transformer-oriented models.

CNN-BiLSTM forecasting architecture



Chronological evaluation and train-only scaling determine validity more than architectural complexity.

Figure 2.1 – Conceptual CNN-BiLSTM forecasting pipeline used to frame the corrected sequence experiments.

2.6 Patch-based temporal representation

PatchTST segments each channel’s lookback into temporal patches and applies a Transformer encoder to the patch tokens [3]. Patching reduces the token count and encourages representation of local subsequences. Channel independence allows the same backbone to process each variable while preserving variable-specific sequences. Reversible Instance Normalization (RevIN) symmetrically removes and restores instance statistics to address temporal distribution shift [12]. It should not be conflated with an arbitrary change of normalization at deployment: a serving transform must be evaluated under the same contract used for the reported claim.

PatchTST was implemented at 24 hours and in long-horizon research notebooks, and it was later evaluated on the common ECL benchmark at 24, 96, 192, 336, and 720 hours. These experiments demonstrate the architecture’s applicability, but final checkpoint selection ultimately depended on the different Low Carbon London household cold-start problem and a minimal user input contract.

2.7 Inverted and dual-stream representations

iTransformer reorients the Transformer so that each variable becomes a token whose embedding summarizes its temporal history. Attention then learns cross-variable relationships, while feed-forward components learn representations within each variable token [4]. This organization is well suited to the multivariate ECL benchmark with 321 clients. In the local benchmark, iTransformer achieves the lowest normalized MAE among the four completed architectures at every recorded horizon.

xPatch is a dual-stream, non-Transformer architecture based on exponential seasonal-trend decomposition. It combines a linear stream with a nonlinear CNN stream and uses patching and channel independence [5]. Its local ECL results are generally close to PatchTST and, at some horizons, achieve lower normalized MAE. The distinction between PatchTST and iTransformer is illustrated in figure 2.2; xPatch should be understood as a complementary decomposition-based competitor rather than another attention architecture.

Two complementary Transformer views of multivariate forecasting

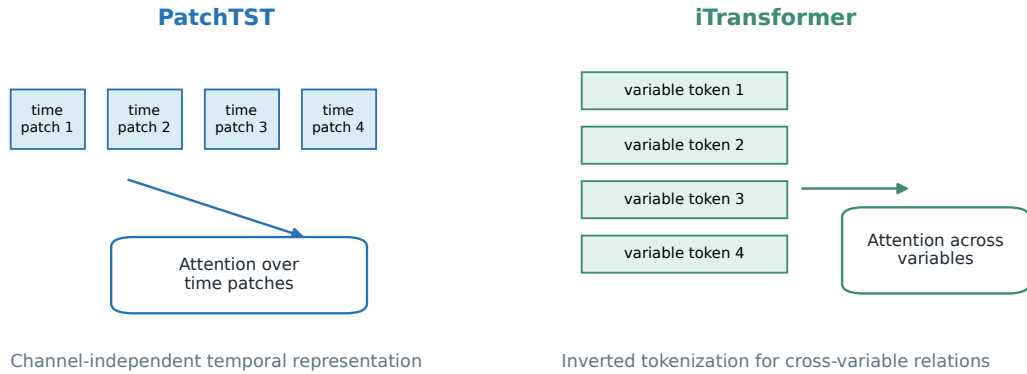


Figure 2.2 – Conceptual comparison between temporal-patch and variable-token representations.

2.8 Recent universal and efficient models

TimeMixer++ proposes multi-resolution time imaging, decomposition, and mixing for several time-series tasks [13]. TimePro addresses multivariate long-term forecasting through variable- and time-aware hyper-states and a Mamba-based design [14]. Both were included in the ECL benchmark notebook’s planned experiment matrix.

Reproducibility conditions differed from the mature baselines. The TimeMixer++ path used a public PyPOTS backbone rather than an author-maintained official training repository. TimePro required compiled selective-scan and DCNv4 operators pinned

to a different PyTorch/CUDA environment. The current evidence tree contains no successful leaderboard rows for either model. They are therefore documented as attempted integrations and software-reproducibility lessons, not as completed local reproductions.

2.9 Global models for many related series

A global model shares parameters across many time series. Its complexity does not grow one-for-one with the number of series, and it can learn structure across a panel without requiring that every series be identical [15]. N-BEATS uses stacks of fully connected residual blocks that produce backcast and forecast components and can be configured for interpretable trend and seasonality [16]. The local product-candidate experiment used a shared global N-BEATS backbone, with optional localization support retained in the worker but disabled in the selected full-profile run because localization did not improve the balanced-run evidence.

Temporal Fusion Transformer (TFT) combines variable selection, gated residual networks, recurrent local processing, and interpretable attention for multi-horizon forecasting [17]. The EnergyAI Global TFT adapts this family to a household input contract consisting of historical energy and derived calendar features. It outputs three quantiles learned through the pinball objective introduced by quantile regression [18] and is trained across households. The artifact manifests document 1,386,262 trainable parameters, a 336-hour lookback, and data-quality gates.

2.10 Pretrained time-series foundation models

Chronos converts scaled time-series values into a representation that can be modeled by a pretrained sequence architecture and returns probabilistic forecasts [6]. Its later official implementation adds direct quantile prediction, longer context, covariates, and parameter-efficient fine-tuning paths [7]. This changes the model-selection tradeoff: a compact task-specific model may remain preferable for latency and memory, while one shared pretrained base can transfer across cadences and datasets when adapted with small LoRA modules. EnergyAI evaluates that proposition under frozen validation and audit gates rather than assuming that pretraining guarantees transfer.

2.11 Ensembles and extraordinary scores

Alghamdi et al. propose a stacked LSTM snapshot ensemble combined with Fast Fourier Transform (FFT) features and a meta-learner and report RMSE 0.020, MAE 0.013, MAPE 0.017, and $R^2 = 0.999$ on IHEPC with weather data [19]. The paper is

relevant because it illustrates how ensemble diversity and frequency-domain features can be combined. It also illustrates why scale, split, target definition, and inverse transformation must accompany extraordinary metrics.

The old PFE folder contains an LSTM classifier, but not evidence of an exact FFT plus snapshot-ensemble plus meta-learner reproduction. This report therefore treats the article as reviewed literature and the local LSTM as a separate experiment. It does not merge their statuses.

2.12 Primary-source synthesis and local evidence boundary

Table 2.1 distinguishes what each primary work contributes from what this PFE actually completed. The table is intentionally paper-level: local metrics are reported later, where dataset, split, units, and artifact status can accompany them.

Table 2.1 – Primary works, methodological relevance, and local status.

Primary work	Contribution relevant to this PFE	Local use and evidence status
DEPM [1]	Hybrid classification pipeline combining neural representations and XGBoost	Starting replication supplied by the supervisor; architecture reconstructed, evaluation claim invalidated by leakage audit
LSTM [10] and GRU [11]	Gated recurrent memory for long dependencies	Standalone and hybrid historical classifiers; later used as causal sequence baselines, not claimed as exact paper reproductions
XGBoost [9]	Scalable regularized tree boosting	Hybrid head and diagnostic baseline in the DEPM phase
EECP-CBL [2]	CNN plus BiLSTM electricity-consumption prediction	Dedicated replication attempt and later family-level ECL baseline; exact published values not claimed as reproduced
RevIN [12]	Reversible instance normalization under distribution shift	Used in patch-oriented research; also clarifies why research/serving normalization equivalence must be tested

Primary work	Contribution relevant to this PFE	Local use and evidence status
PatchTST [3]	Channel-independent temporal patch tokens	Completed ECL rows; historical product research archived
iTransformer [4]	Variable-token attention for multivariate forecasting	Completed ECL rows and best stored normalized MAE at every completed horizon
xPatch [5]	Seasonal-trend decomposition with linear and nonlinear patch streams	Completed ECL rows; treated as a separate non-Transformer competitor
N-BEATS [16]	Residual backcast/forecast basis expansion	Completed LCL candidate rows; not the selected packaged model
TFT [17]	Gating, variable selection, recurrent context, attention, and multi-horizon quantiles	Completed LCL 24 h and 168 h checkpoints; packaged and separately evaluated under serving normalization on the frozen LCL cohort; new-site transfer remains unvalidated
Chronos / Chronos-2 [6], [7]	Pretrained probabilistic forecasting with parameter-efficient adaptation	Selected, transfer-audited, and deployed in the repository for 30 daily targets; hourly challengers packaged but not activated
TimeMixer++ [13] and TimePro [14]	Recent multi-resolution and efficient multivariate designs	Integration attempted; dependency/provenance barriers prevented completed local leaderboard rows
Probabilistic and conformal forecasting [20], [21]	Calibration, sharpness, and post-hoc interval validity	TFT intervals are diagnosed; the month release applies a frozen residual-based conformal correction

2.13 Methodological risks across model families

2.13.1 Temporal leakage

Leakage occurs whenever information unavailable at prediction time influences model fitting or evaluation [22]. Direct leakage includes the target or a deterministic transformation of it. Proxy leakage occurs when other variables reconstruct the target through

domain relations. Preprocessing leakage occurs when the scaler, dimensionality reduction, feature selection, or imputation learns from validation or test observations. Window leakage occurs when centered or non-shifted rolling statistics include the prediction point.

2.13.2 Inappropriate splitting

Random cross-validation assumes exchangeability. Consecutive electricity measurements violate this assumption because adjacent values share regime, weather, occupancy, and season. Time-aware evaluation must preserve ordering or use rolling-origin procedures. Even when some autoregressive cross-validation schemes can be valid under specific assumptions [23], a deployment-oriented split must imitate what the application will know at prediction time.

2.13.3 Metric aggregation

Global R^2 can be dominated by high-load clients, whereas macro household R^2 gives each client equal weight. MAPE is unstable near zero. sMAPE reduces but does not eliminate that problem. RMSE emphasizes large errors; MAE is more robust. MASE compares performance with a seasonal rule and is useful across scales. No single metric is sufficient; the EnergyAI selection considers macro MAE, median, p90, seasonal improvement, proportion of households beating baseline, bias, and inference cost.

2.13.4 Calibration, drift, and data responsibility

Probabilistic forecasts should be evaluated for both calibration and sharpness rather than by point error alone [20]. A nominal central 80% interval should therefore be accompanied by empirical coverage and interval width. Under changing load regimes, concept drift can alter the relationship between recent context and future demand [24]; detection and governed response are deployment responsibilities, not automatic consequences of a good static test score. Finally, fine-grained smart-meter traces can reveal household routines, so ownership scoping, minimized exports, and bounded retention are part of model-system quality [25].

Table 2.2 – Model families and their role in this PFE.

Family	Main strength	Main risk	PFE status
Seasonal naive	Transparent, strong periodic reference	Cannot adapt to changing regimes	Mandatory baseline
Linear/tree ML	Fast, interpretable engineered features	Leakage through feature construction	Diagnostic and comparison
LSTM/GRU	Sequential state and nonlinear dynamics	Random windows can inflate performance	Historical and corrected experiments
CNN-BiLSTM	Local motifs plus sequence context	Complexity without causal validation	Corrected baseline and ECL benchmark
PatchTST	Efficient temporal patches	Benchmark-to-product task mismatch	Completed benchmark
iTransformer	Cross-variable relations	Global metrics dominated by large clients	Completed ECL benchmark
xPatch	Decomposition with linear/nonlinear streams	Recent implementation sensitivity	Completed ECL benchmark
TimePro / TimeMixer++	Recent efficient designs	Reproduction uncertainty	Attempted; no completed metrics
N-BEATS	Strong global residual backbone	Point forecast without native uncertainty in local setup	Product fall-back/experimental
TFT	Multi-horizon quantiles and gating	More complex artifact and calibration needs	Selected and packaged at 24 h and 168 h
Chronos-2 plus LoRA	Cross-dataset pretrained representation and small adapters	Base memory, startup latency, and transfer sensitivity	Active at 30 daily targets; hourly challengers retained

The review leads to a clear experimental principle: the model family is only one component of the claim. Target construction, causal information, split policy, preprocessing scope, baseline, aggregation, and artifact provenance are equally necessary. The next chapter applies this principle to the initial DEPM replication.

Critical Replication of the Deep Energy Predictor Model

3.1 Origin and objective of the replication

The initial research article, provided as the starting point of the PFE, was the DEPM study by Ragupathi et al. [1]. The paper frames the problem as electricity-consumption prediction and combines XGBoost classification, a DNN, and a cascaded residual network. It reports 98% accuracy, precision 0.97, recall 0.99, and F1-score 0.98 on the Individual Household Electric Power Consumption (IHEPC) dataset.

The replication objective was initially literal: reconstruct preprocessing, feature extraction, neural components, and classifier; reproduce the announced metrics; and study architectural variants. The old PFE workspace contains the source PDF, a French summary, multiple DEPM notebooks, feature-ablation and comparison notebooks, and DNN/ResNet/XGBoost artifacts. The Git repository later added a full replication, a no-leakage variant, recurrent variants, and a Steel Industry cross-dataset experiment.

The work reproduced accuracy around 99.29% in an early local pipeline. That number is historically important because it explains why the project initially appeared successful. It is not a final result. The subsequent audit showed that the target and split allowed models to infer the label without learning a deployable temporal relationship.

3.2 Reconstructed architecture

The reconstructed flow is summarized in figure 3.1. Numeric features were standardized and reduced with PCA. A cascaded residual network and a DNN were used as nonlinear feature extractors, while XGBoost produced the final binary class. Later local variants replaced the DNN with LSTM, BiLSTM, GRU, or BiGRU representations and also evaluated each neural model independently.

Simplified DEPM architecture reconstructed during the initial phase



The architecture was reproducible; the central problem was the target and validation protocol.

Figure 3.1 – Simplified DEPM architecture reconstructed from the article and local notebooks.

The architecture itself could be implemented. The scientific issue concerned what information entered it and what the evaluation score represented. This distinction matters: a replication can correctly reproduce code behavior while exposing that the original experimental description is insufficient for the claimed use case.

3.3 Target construction and direct leakage

Several early notebooks constructed a binary variable named `consumption_class` from `Global_active_power`. Feature selection then removed only the new class column, leaving `Global_active_power` in the input matrix. The classifier was therefore asked to predict a thresholded transformation of a value it already received. Under that definition, high accuracy is expected and does not demonstrate forecasting.

Later notebooks excluded the raw active-power column, but some retained rolling or interaction features calculated from the same signal at the current time. If a rolling mean for time t includes y_t , it remains target leakage even though the original column name is absent. The correct causal feature is based only on $y_{t-k:t-1}$ when predicting y_t or later.

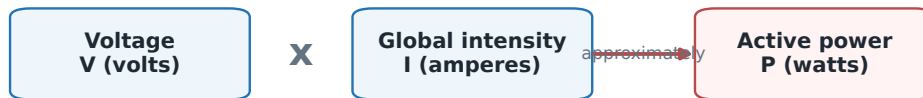
3.4 Physical proxy leakage

The IHEPC variables include voltage and global intensity. In an electrical circuit, active power can be approximated from voltage and current, with power factor and phase effects determining the exact relationship. At household aggregate level, the relation

$$P \approx V \times I \quad (3.1)$$

is sufficiently strong to make the label derived from active power largely reconstructible from simultaneous voltage and current. Some local feature pipelines made this relation even more explicit through `voltage_x_intensity`.

A physically reconstructible target creates proxy leakage



If the class label is derived from active power, V and I can nearly reconstruct it.

A simple classifier can therefore score highly without learning temporal demand dynamics.

Figure 3.2 – Physical proxy mechanism that made the high/low active-power class reconstructible.

This is a domain-specific form of leakage. A feature may not be named after the target, yet it can encode the same quantity. Domain knowledge is therefore a necessary part of the data audit. The presence of a physical proxy does not make voltage or current universally unusable; they may be valid historical predictors when appropriately lagged and when the deployment system actually observes them. The invalid condition is using simultaneous values to predict a target that is algebraically related to them.

3.5 Preprocessing leakage

The historical workflow fitted both the standard scaler and the PCA transformation to the complete feature matrix before creating the train-test split. In that order, the mean, variance, and principal directions include information from the held-out observations. The classifier does not receive test labels directly, but the representation has been optimized using the test distribution.

The corrected order is:

1. define chronological train, validation, and test indices;
2. fit the scaler on the training region;
3. transform validation and test with the frozen training scaler;
4. fit any dimensionality reduction or feature selector on the transformed training region;
5. apply the fitted transformation unchanged to later regions.

3.6 Random temporal splitting

The early experiments used random stratified splitting. Adjacent electricity measurements can be nearly identical, especially at minute-level sampling. Random allocation places observations from the same day, household regime, and local trend in both train and test. Sequence construction does not automatically solve the problem: if overlapping windows are randomly split after creation, their inputs may share most time steps.

Chronological separation is necessary for the target product question. Training data must precede validation data, which must precede test data. A clean boundary may allow the first validation window to use earlier training observations as historical context; it must not allow a training target after the validation start or a scaler fitted on later data.

3.7 Classification is not forecasting

The DEPM evaluation answers whether a contemporaneous measurement is assigned to a high or low class. EnergyAI must estimate a sequence of future energy values. The two tasks differ in output, loss, evaluation, and usefulness:

- classification outputs a probability or category; forecasting outputs an ordered trajectory;
- classification accuracy ignores the magnitude of an energy error;
- a class threshold can hide large within-class deviations;
- operational horizons require that all inputs be known before the first predicted timestamp;
- physical-unit MAE and RMSE are needed for cost and load interpretation.

The article’s class-threshold definition could not be reconstructed with sufficient confidence from the local source material. This ambiguity is reported as a reproducibility limitation, not as evidence of intent.

3.8 Diagnostic logistic regression

A simple logistic regression achieved 99.4% accuracy when applied to the leaky feature set. The result is more informative than another complex-model comparison. It shows that the classification boundary is almost directly encoded in the predictors. If a linear separator equals or exceeds the reported hybrid score, the high accuracy cannot be used to demonstrate the necessity of cascaded ResNet, DNN, or XGBoost components.

The diagnostic should not be misunderstood as a candidate energy forecaster. It was intentionally trained on the suspect classification formulation to test whether the task

was trivial. Its role is analogous to a unit test for leakage.

3.9 Historical corrected variants

The later archive contains results for standalone DNN, LSTM, BiLSTM, GRU, and BiGRU classifiers and their DEPM-XGBoost hybrids after removing important direct-leak features. Recorded accuracies range from 0.8852 to 0.8959. Table 3.1 reproduces selected values from `models/archive/results.csv`.

Table 3.1 – Selected leakage-reduced classification results. These remain non-forecasting results.

Model	Accuracy	Precision	Recall	F1
DNN	0.8953	0.9172	0.8691	0.8925
LSTM	0.8904	0.9314	0.8428	0.8849
BiLSTM	0.8885	0.9384	0.8315	0.8817
DEPM-DNN	0.8959	0.9314	0.8548	0.8914
DEPM-LSTM	0.8923	0.9204	0.8589	0.8886
DEPM-BiGRU	0.8852	0.9094	0.8557	0.8817

These lower results are scientifically healthier because they reflect removal of the most direct answer channels. They also show that the hybrid does not dominate the standalone DNN by a meaningful margin. Nonetheless, the random-classification framing and any remaining preprocessing caveats prevent use of this table as evidence for future continuous forecasting.

3.10 Documentary inconsistencies and neutral reporting

The audit identified additional elements that could not be reconciled from the article alone: the exact class threshold, the apparent total in a published confusion matrix compared with the stated dataset size, the evaluation subset used for that figure, and the scale on which some error metrics were reported. Some terminology also appears unrelated to electricity prediction. The correct academic conclusion is limited: the article does not provide enough information to reproduce these elements unambiguously.

This report does not accuse the authors of cheating, fabrication, or plagiarism. Such claims would require a separate formal investigation. Instead, it states observable discrepancies, their effect on reproducibility, and the evidence available locally.

3.11 Scientific decision to pivot

Figure 3.3 summarizes the change in protocol. The pivot did not discard the initial work; it converted it into a methodological case study and established stricter requirements for every later experiment.

Why the initial score did not measure future forecasting ability

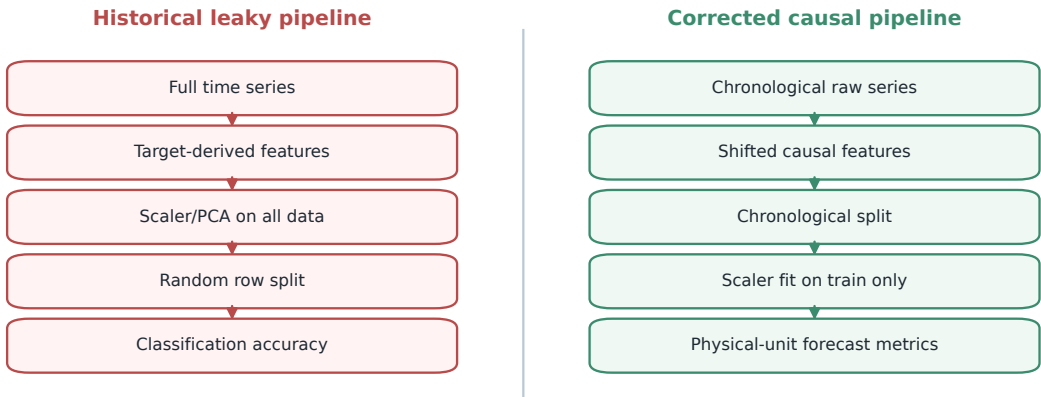


Figure 3.3 – Contrast between the historical leaky pipeline and the corrected causal pipeline.

The decision was to predict continuous energy at explicit future horizons, use lagged inputs only, compare with seasonal rules, preserve physical units, separate model selection from final testing, and require a stored source for every published metric. The following chapter formalizes that protocol.

Reliable Experimental Protocol

4.1 Continuous multi-horizon formulation

The corrected task estimates future hourly energy rather than a contemporaneous high/low class. For an origin t , lookback L , and horizon H , the input contains observations $y_{t-L+1}, \dots, y_t$ and the target contains $y_{t+1}, \dots, y_{t+H}$. Calendar features for future timestamps are permitted because hour, weekday, month, workday status, and holiday flags are known before inference. Future measured consumption or future weather observations are not permitted unless replaced by forecasts available at the same origin.

EnergyAI uses two cadence-specific contracts. The hourly TFTs use $L = 336$ hourly values and $H \in \{24, 168\}$ hourly targets. The month model aggregates meter intervals into rolling 24-hour energy totals, accepts up to 365 daily values with a minimum of 270 valid days, and directly emits 30 daily targets. The API retains 720 as a duration identifier for compatibility while returning the authoritative target count and interval explicitly. Direct multi-output forecasting avoids the error accumulation of recursively feeding each prediction back as the next input.

4.2 Causal windows

A window is causal when every input could have been observed at its forecast origin. Lag features use y_{t-k} with $k \geq 0$; rolling statistics end at t when predicting $t + 1$ or later. If the target begins at t , the rolling window must end at $t - 1$. This indexing convention is stated explicitly in each notebook or worker.

Missing inputs are not silently imputed from future observations. The research preparation performs bounded causal forward filling after hourly aggregation and excludes evaluation windows that remain non-finite. The serving manifests instead require at least 95% coverage, no consecutive gap longer than three hours, and finite model inputs. These policies share a causal intent but are not identical; the report therefore treats preprocessing equivalence as a validation requirement, not as an established property.

4.3 Chronological separation

The ECL benchmark reconstructs a common 321-client hourly matrix and uses 18,317 rows for training, 2,633 for validation, and 5,261 for testing. The Low Carbon London workflow first selects 2,500 eligible 2013 households, assigns 2,000 to the known cohort and 500 to the cold-start cohort, and then applies one common hourly chronology to every series. Indices 0–6,131 form training (1 January 00:00 through 13 September 11:00 UTC), 6,132–7,007 form validation (13 September 12:00 through 19 October 23:00 UTC), and 7,008–8,759 form test (20 October 00:00 through 31 December 23:00 UTC). Cold-start households contribute no gradient updates. Their test results therefore measure both later time and transfer to unseen households.

Low Carbon London: one shared 2013 chronology for every household

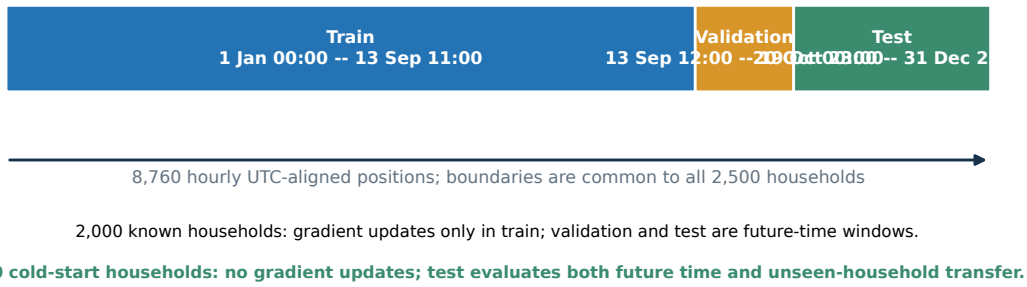


Figure 4.1 – Chronological split used to prevent future observations from influencing model fitting.

Validation guides early stopping and architecture choice. The test region must not be consulted repeatedly while tuning. Once the final configuration is selected, a final test evaluation is recorded. If multiple seeds are run, the selection rule and aggregation must be declared before inspecting the final cohort.

4.4 Resampling, units, and time zones

The LCL source field is named `energy(kWh/hh)`: each value is energy consumed during a half-hour interval, not an instantaneous power measurement. The preparation parses timestamps as UTC, converts them to timezone-naive UTC indices, aggregates duplicate half-hours by timestamp, and sums the two half-hour energy values into each hourly total. Consequently, an hourly target and its MAE are expressed in kWh for an hourly interval. The report does not reinterpret these values as kW. Daily targets are sums in kWh and require at least 22 finite hourly values.

All 2013 households are reindexed to the same 8,760-position UTC calendar before the shared split is applied. This avoids a local-clock duplicate hour in autumn and a missing hour in spring: daylight-saving transitions do not create ambiguous positions in the research matrix. The product accepts timezone-aware site settings and normalizes stored timestamps through its ingestion contract. The supplementary serving-normalization evaluation reuses the preserved calendar rows of the frozen LCL samples, so it isolates the scaler change without claiming that arbitrary site time zones are equivalent to the LCL preparation.

4.5 Train-only transformations

For a training variable x , standardization uses

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}. \quad (4.1)$$

The same μ_{train} and σ_{train} transform validation and test data. Predictions are converted back with

$$\hat{x} = \hat{z}\sigma_{\text{train}} + \mu_{\text{train}}. \quad (4.2)$$

The ECL benchmark uses per-client means and standard deviations learned from its training rows. The LCL training worker also stores one mean and standard deviation per household, calculated only from positions before the common training boundary. This includes each cold-start household’s historical training segment: those values define its evaluation scale but are never used for gradient updates.

EnergyAI serving does something different. For a new site, it calculates mean and standard deviation from the latest accepted 336-hour input window, normalizes that window, and inversely transforms outputs with those same rolling statistics. A supplementary inference-only evaluation has now rerun the exact frozen LCL cold-start households, origins, targets, seasonal baselines, and checkpoints with this rolling transform and the production non-negative, ordered-quantile output handling. This closes the transform-specific evidence gap on the preserved London cohort. It does not convert the original research metrics into serving metrics, nor does it validate transfer to a new Moroccan site.

Table 4.1 – Normalization contracts and what the evidence supports.

Context	Statistics used	Supported interpretation
ECL benchmark	Per client, fitted on benchmark training rows	Valid for stored ECL benchmark rows only
LCL known-household evaluation	Per household, fitted on that household’s common training segment	Valid for future-time windows of known London households
LCL cold-start evaluation	Per cold household, fitted on its historical pre-validation segment; no gradient use	Valid for London household transfer when historical scaling statistics are available
EnergyAI serving	Per site, calculated from the latest accepted 336-hour window	Implemented and evaluated on the preserved frozen LCL cohort; new-site and Moroccan transfer remain unvalidated

4.6 Datasets and their roles

Table 4.2 – Datasets used at different stages of the research.

Dataset	Granularity / entity	Role	Important caveat
IHEPC [26]	One French household, minute-level source	DEPM replication, leakage audit, CNN-BiLSTM pivot	A single household does not demonstrate cross-household generalization
Steel Industry	Industrial consumption records	Cross-dataset DEPM transfer experiment	Different domain and target distribution
ECL [27]	321-client hourly benchmark after deterministic preparation	Common multivariate deep-model benchmark	Multivariate matrix task differs from one new user’s CSV
Low Carbon London [28]	Thousands of smart-meter households	Global-model selection and cold-start evaluation	Geographic and behavioral transfer to Morocco remains unverified

No dataset is treated as interchangeable. IHEPC is valuable for replicating the article that motivated the PFE. ECL supports comparison with modern multivariate forecasting literature. Low Carbon London is closest to the product’s many-household transfer problem and is therefore the source of the final checkpoint-selection metrics.

4.7 Baselines

At minimum, each horizon should be compared with:

- persistence, which repeats the most recent observation or profile;
- daily seasonal naive, which reuses the corresponding hour from the previous day;
- weekly seasonal naive when enough history exists;
- a simple linear or tree-based model when engineered covariates are available.

The artifact manifests expose comparison with a seasonal-naive baseline. The cold-start 24-hour baseline macro MAE is 0.251540 kWh per hourly interval; the 168-hour value is 0.249055 kWh per hourly interval. Model selection requires a positive and sufficiently broad improvement over this reference.

4.8 Metrics

For n target points, MAE is

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (4.3)$$

and Root Mean Squared Error (RMSE) is

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (4.4)$$

MAE is the primary product-selection metric because it is robust and directly interpretable after inverse transformation. RMSE highlights large misses. Bias, $n^{-1} \sum_i (\hat{y}_i - y_i)$, identifies systematic over- or under-prediction.

For quantile q , TFT minimizes the pinball loss

$$\rho_q(y - \hat{y}_q) = \max((q - 1)(y - \hat{y}_q), q(y - \hat{y}_q)). \quad (4.5)$$

The local model averages this loss over $q \in \{0.1, 0.5, 0.9\}$, samples, and lead times. The 0.5 output is the median point forecast. The 0.1 and 0.9 outputs define a nominal central 80% interval, whose empirical coverage and mean width are reported separately rather than assumed from the quantile labels [18], [20].

Symmetric Mean Absolute Percentage Error (sMAPE) is computed as

$$\text{sMAPE} = \frac{100}{n} \sum_i \frac{2|y_i - \hat{y}_i|}{|y_i| + |\hat{y}_i| + \epsilon}. \quad (4.6)$$

It remains sensitive near zero. Mean Absolute Scaled Error (MASE) divides MAE by an in-sample seasonal-naive error and provides a scale-relative view. R^2 is useful but depends on aggregation: global flattening can overweight high-load clients, whereas macro R^2 averages client-level coefficients. Negative macro R^2 means the average client forecast is worse than a mean-based reference under that definition and must not be hidden behind a favorable global score.

4.9 Known-household and cold-start evaluation

Known-household metrics evaluate future windows for households represented during training. Cold-start metrics evaluate households completely held out from gradient updates. For a platform accepting a new user’s recent history, cold-start evidence is the stronger selection criterion. EnergyAI therefore reports cold-start macro MAE, median household MAE, p90 household MAE, macro and global R^2 , seasonal improvement, and household win rate.

Macro aggregation first calculates a metric per household and then averages those values. It prevents a few high-consumption households from dominating the decision. The p90 value shows the error below which 90% of households fall and makes the long tail visible.

4.10 Reproducibility controls

Each experiment should record the dataset transformation, split boundaries, lookback, horizon, seed, batch size, epochs, early-stopping rule, optimizer, parameter count, training time, checkpoint, metric file, and software source. Production artifacts add:

- a versioned artifact-contract number;
- a checkpoint filename and SHA-256 digest;
- input and output units;
- data-quality gates;
- output quantiles and expected shape;
- architecture metadata;
- limitations and smoke-reference points.

The 24-hour checkpoint digest is `60fedcde...bc5a04d`; the 168-hour digest is `80af16b2...95f93`. The full hashes are stored in the artifact hash inventory under

report/evidence. The report registry also marks each artifact as historical, invalidated, benchmark, candidate, experimental, or a packaged product capability with explicit limitations.

4.11 Promotion discipline

Evidence-gated model promotion

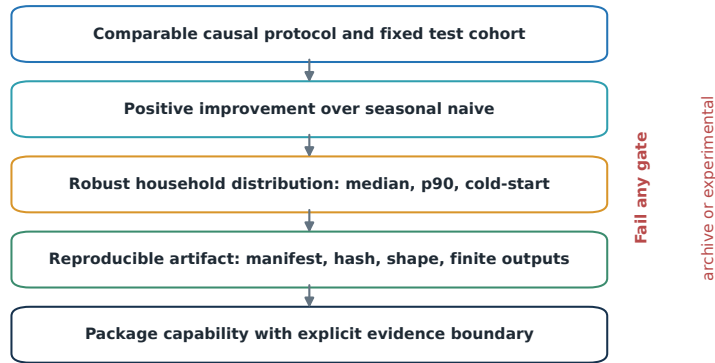


Figure 4.2 – Evidence gates used to separate an experiment from an exposed product capability.

Completion of training is not sufficient for promotion. The run must use a comparable causal protocol, beat frozen comparators, remain robust across the evaluated distribution, load strictly from a reproducible artifact, and produce finite outputs of the expected shape. The month study froze selection on Tetouan before opening Portugal and southern-Morocco diagnostics; the latter negative transfer result is retained as a limitation. The hourly Chronos challengers passed their London accuracy gates but were not activated because their small improvement did not yet satisfy the broader serving tradeoff. TimePro and TimeMixer++ are still not assigned local performance claims without successful metric files.

Experiments, Benchmarks, and Final Model Selection

5.1 Experimental narrative and evidence status

The experimental history is broader than the final product. This chapter preserves that breadth while separating scientific status. Historical DEPM and LSTM classifications explain the audit; corrected IHEPC and Steel Industry notebooks show the pivot; ECL provides a common modern-architecture benchmark; and Low Carbon London supports product selection. A model is described as completed only when a notebook output, leaderboard row, manifest, or checkpoint-backed metric file exists.

5.2 Corrected DEPM and Steel Industry experiments

The DEPM phase evaluated standalone DNN, LSTM, BiLSTM, GRU, and BiGRU models and corresponding XGBoost hybrids. After removal of the most direct target channels, classification accuracy fell to approximately 0.885-0.896, as reported in table 3.1. This result is interpreted as evidence that the historical 99% score depended on target reconstruction. It does not establish a useful continuous forecaster.

The Steel Industry notebook tested whether the DEPM-style pipeline transferred to a different energy dataset. This was a valuable external-generalization attempt because a method tied to IHEPC variable identities would be fragile. However, the dataset represents industrial rather than household demand, and any class-based result remains methodologically different from the product target. The experiment is retained in the notebook and Git catalog without promoting unsupported numbers into the report body.

5.3 CNN-BiLSTM and the regression pivot

The EECF-CBL work of Le et al. motivated a continuous CNN-BiLSTM replication [2]. The local protocol used chronological splitting, fit its scaler on training data, evaluated without shuffling, and converted errors to the original scale. This experiment

established the mechanics that the DEPM work lacked: sequence-to-future mapping, physical-unit error, and explicit horizon handling.

The exact architecture and dataset variants in the paper were not treated as automatically reproduced. A local model can belong to the same family without matching the authors’ code, preprocessing, and data subsets. The report therefore uses “CNN-BiLSTM baseline” for the later ECL implementation and “EECP-CBL replication attempt” only for the corresponding dedicated notebook.

5.4 PatchTST and long-horizon research

PatchTST experiments introduced direct multi-step forecasting with patch tokens, channel independence, and reversible normalization. Research checkpoints existed for 24, 168, and 720-hour model families at the v0.1.0 tag, alongside advanced PatchTST and iTransformer artifacts. These assets document the evolution of the modeling work. They remain research archives rather than being confused with the later active TFT and Chronos-2 releases.

5.5 ECL deep-model benchmark

The ECL benchmark used a common data matrix, chronological split, train-only per-client standardization, fixed seed 2026, early stopping, and horizon-specific batch sizes. Completed leaderboard rows exist for iTransformer, PatchTST, xPatch, and CNN-BiLSTM at 24, 96, 192, 336, and 720 hours. The notebook planned TimePro and TimeMixer++, but no successful leaderboard rows exist for them.

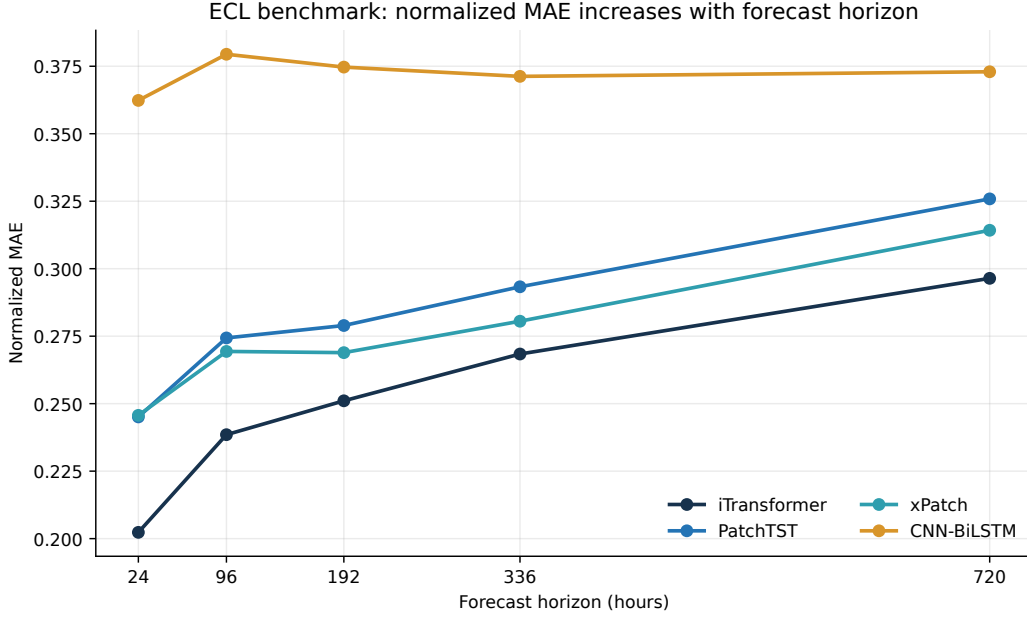


Figure 5.1 – Normalized MAE from the stored ECL leaderboard. The chart includes completed runs only.

Table 5.1 – Selected ECL results from the completed local benchmark.

Horizon	Model	Norm. MSE	Norm. MAE	Macro R^2	Parameters
24	iTransformer	0.1076	0.2023	0.8723	6,374,936
24	PatchTST	0.1467	0.2451	0.8304	436,632
24	xPatch	0.1543	0.2457	0.8223	32,074
24	CNN-BiLSTM	0.2782	0.3623	0.4971	454,129
720	iTransformer	0.2034	0.2964	0.7575	6,731,984
720	PatchTST	0.2390	0.3259	0.7191	1,506,384
720	xPatch	0.2309	0.3142	0.7255	3,942,898
720	CNN-BiLSTM	0.2883	0.3730	0.4099	4,763,761

iTransformer has the lowest normalized MAE among the completed models at every stored horizon. At 24 hours it obtains 0.2023, compared with 0.2451 for PatchTST, 0.2457 for xPatch, and 0.3623 for CNN-BiLSTM. At 720 hours, its normalized MAE is 0.2964. The performance gap does not alone make iTransformer the product model: ECL is a simultaneous 321-client benchmark, whereas EnergyAI accepts a single site’s timestamp-energy history and must generalize to a new household.

The physical-unit MAE column in the ECL leaderboard is aggregated across the reconstructed client matrix and reaches tens of kWh. It must not be compared directly with per-household LCL MAE. Normalized ECL metrics are used for within-benchmark model comparison only.

5.6 Incomplete recent-model attempts

The benchmark worker isolated each third-party repository in a separate process because several projects use conflicting top-level module names such as `models` and `layers`. TimePro additionally required compiled CUDA extensions against versions different from the current Kaggle image. TimeMixer++ used a PyPOTS implementation because a stable author-maintained training repository was not available in the reviewed sources.

The absence of successful rows is a result in itself: modern-paper reproduction depends on environment, dependency, and implementation provenance. These attempts are documented in the notebook catalog and state of the art, but excluded from rankings and conclusions.

5.7 Low Carbon London global forecasting

The final selection problem used a global univariate household formulation. Each sample contains one household’s historical energy and target, while batches mix windows from many households. Calendar features are derived from timestamps. The full-profile experiment used 2,500 eligible 2013 households: 2,000 known households and 500 cold-start households. With 48 sampled windows per known household, one nominal training epoch contains 96,000 dynamically selected windows. Validation and test use 24 fixed, evenly spaced candidate origins per household; windows containing non-finite required values are omitted. The selected candidates were Global TFT and Global N-BEATS for day and week horizons, plus Global N-BEATS and MultiCycleNet-adapted for monthly daily totals.

5.7.1 Frozen TFT configuration

Table 5.2 – Global TFT configuration recorded by the full-profile worker and notebook.

Property	24-hour run	168-hour run
Lookback / horizon	336 / 24 hourly values	336 / 168 hourly values
Quantiles	0.1, 0.5, 0.9	0.1, 0.5, 0.9
Hidden size / heads / dropout	128 / 4 / 0.1	128 / 4 / 0.1
Optimizer	AdamW, initial learning rate 3×10^{-4} , weight decay 10^{-4}	Same
Schedule and stabilization	Linear warm-up then cosine decay; gradient-norm clip 1.0; float16 automatic mixed precision	Same
Batch / nominal windows per epoch	128 / 96,000	128 / 96,000
Early stopping	Maximum 15 epochs, patience 5	Maximum 15 epochs, patience 5
Completed epochs / best validation loss	15 / 0.141531	14 / 0.148559
Trainable parameters	1,386,262	1,386,262

The notebook records Python 3.12.13, PyTorch 2.10.0 with CUDA 12.8, and two Tesla T4 GPUs with approximately 14.6 GiB each. Selected runs were scheduled in parallel across the two devices. The 24-hour TFT recorded 1,356.70 seconds of training and 1,372.01 seconds of total wall time; the corresponding 168-hour values were 1,666.74 and 1,681.47 seconds. The experiment seed is 2026, but the dynamic training dataset mixes that seed with a clock-derived term and enables cuDNN benchmarking. The frozen files are reproducible artifacts, while exact bit-for-bit retraining is not claimed.

5.7.2 Day-ahead results

On 500 cold-start households and 11,871 valid evaluation windows, Global TFT achieves macro MAE 0.184928 kWh per hourly interval, median household MAE 0.142566, p90 household MAE 0.362807, macro RMSE 0.328317, macro $R^2 = 0.2704$, and global $R^2 = 0.6094$. The seasonal-naive macro MAE is 0.251540. The improvement is 26.48%, and 495 of 500 households (99.0%) beat the baseline.

Global N-BEATS achieves macro MAE 0.195531, a 22.27% improvement, with 97.4% of households beating seasonal naive. Although its macro RMSE is slightly lower than TFT's, TFT has better primary macro MAE, median, p90, macro R^2 , and household win rate. It was selected as primary, with N-BEATS retained as a research fallback.

5.7.3 Week-ahead results

At 168 hours, 499 cold-start households contribute 11,830 valid evaluation windows; one cold household has no complete eligible window. Global TFT cold-start macro MAE is 0.197322 kWh per hourly interval, median 0.149232, p90 0.389872, and macro RMSE 0.341090. It improves on the 0.249055 seasonal baseline by 20.77%, and 491 of 499 households (98.40%) win. Global N-BEATS obtains macro MAE 0.218623, a 12.22% improvement, and an 88.98% household win rate. TFT again provides the stronger primary-error and distributional evidence.

Table 5.3 – Frozen LCL cold-start candidate comparison (MAE in kWh per hourly interval).

Horizon	Model	Macro MAE	Seasonal MAE	Improvement	Win rate
24 h	Global TFT	0.184928	0.251540	26.48%	99.00%
24 h	Global N-BEATS	0.195531	0.251540	22.27%	97.40%
168 h	Global TFT	0.197322	0.249055	20.77%	98.40%
168 h	Global N-BEATS	0.218623	0.249055	12.22%	88.98%

Table 5.4 – Exact frozen cold-start TFT metrics by horizon.

Metric	24 h	168 h
Evaluated households	500	499
Evaluation windows	11,871	11,830
Macro MAE (kWh)	0.184928	0.197322
Median household MAE (kWh)	0.142566	0.149232
p75 household MAE (kWh)	0.234253	0.240102
p90 household MAE (kWh)	0.362807	0.389872
Macro RMSE (kWh)	0.328317	0.341090
Macro sMAPE (%)	38.4987	40.8779
Macro bias (kWh)	-0.074996	-0.076080
Macro R^2	0.270362	0.216235
Median R^2	0.262786	0.230578
Global R^2	0.609431	0.548724

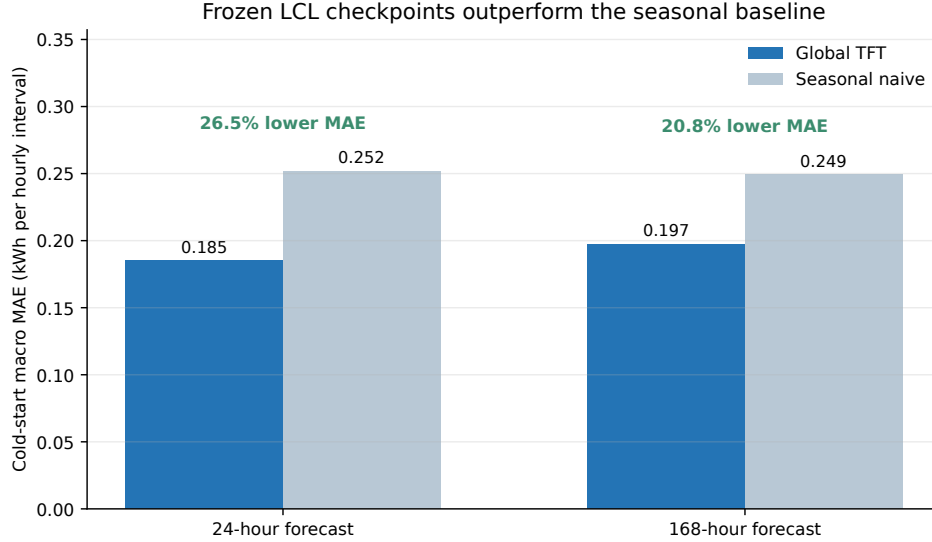


Figure 5.2 – Cold-start Global TFT MAE compared with the active manifests’ seasonal baselines.

5.7.4 Known versus cold-start households

The day-ahead TFT known-household macro MAE is 0.183386, close to the 0.184928 cold-start value. At 168 hours the known value is 0.194966 versus 0.197322 for cold-start. The small gaps are encouraging because the model does not appear to depend strongly on household identities seen during training. They do not establish universal transfer: both cohorts come from the same London research source.

5.7.5 Post-hoc checkpoint diagnostics

The original run metrics retain the authoritative headline values. To examine the quantile outputs that those JSON summaries did not aggregate, a reproducible script reloaded the two frozen `best.pt` checkpoints, rebuilt exactly the cold-start fixed-origin samples, used the original per-household training-segment scalers, and ran float16 inference. The recomputed macro MAE differs from the frozen value by only 3.50×10^{-7} kWh at 24 hours and 1.35×10^{-7} kWh at 168 hours; it also reproduces the exact 495/500 and 491/499 household win counts. These checks guard against accidentally analysing a different cohort or transform.

Table 5.5 – Post-hoc quantile diagnostics from the frozen TFT checkpoints.

Diagnostic	24 h	168 h
Nominal central-interval coverage	80.00%	80.00%
Empirical central-interval coverage	80.7963%	78.3518%
Mean central-interval width (kWh)	0.559770	0.585659
q10 pinball loss (kWh)	0.029654	0.031180
q50 pinball loss (kWh)	0.092568	0.098576
q90 pinball loss (kWh)	0.064440	0.068650
Mean pinball loss (kWh)	0.062221	0.066135
Quantile-crossing rate	0.01895%	0.02742%

The 24-hour central interval is close to its nominal coverage on the frozen London cohort; the 168-hour interval undercovers by approximately 1.65 percentage points and is wider. The asymmetric q10 and q90 pinball losses are consistent with the negative median-forecast bias in table 5.4. These are descriptive test-cohort diagnostics, not a calibrated guarantee for a new site.

5.7.6 Serving-normalization evaluation

The preserved experiment directory was sufficient for an exact inference-only comparison without the original large CSV source. It contains the complete 2,500-household hourly matrix, ordered identities, disjoint known and cold-start assignments, common split boundaries, nine calendar features, and both frozen checkpoints. The 24 fixed test origins, 336-hour histories, future targets, and 168-hour seasonal-naive profiles were reconstructed from these arrays. The research checkpoints and packaged production copies have identical SHA-256 digests for both horizons.

The supplementary run replaced only the research household scaler with the product transformation: the mean and population standard deviation of each current 336-hour history. It then used the production inverse transform, non-negative clamp, and per-step quantile ordering. No training, fine-tuning, checkpoint replacement, or data download occurred. Table 5.6 keeps the original research results and the new serving-transform evidence in separate columns.

On this preserved London cohort, rolling normalization yields slightly lower macro MAE and macro RMSE at both horizons while retaining the same seasonal baseline. The 24-hour interval is slightly wider and covers 81.0108%; the 168-hour interval is also wider but covers 78.0556%. The serving crossing rate is zero by construction because the production path orders the three outputs at every step. These supplementary results close the identified scaler mismatch for this fixed cohort only. They do not replace the

frozen research headlines, establish Moroccan transfer, or provide per-site calibration guarantees.

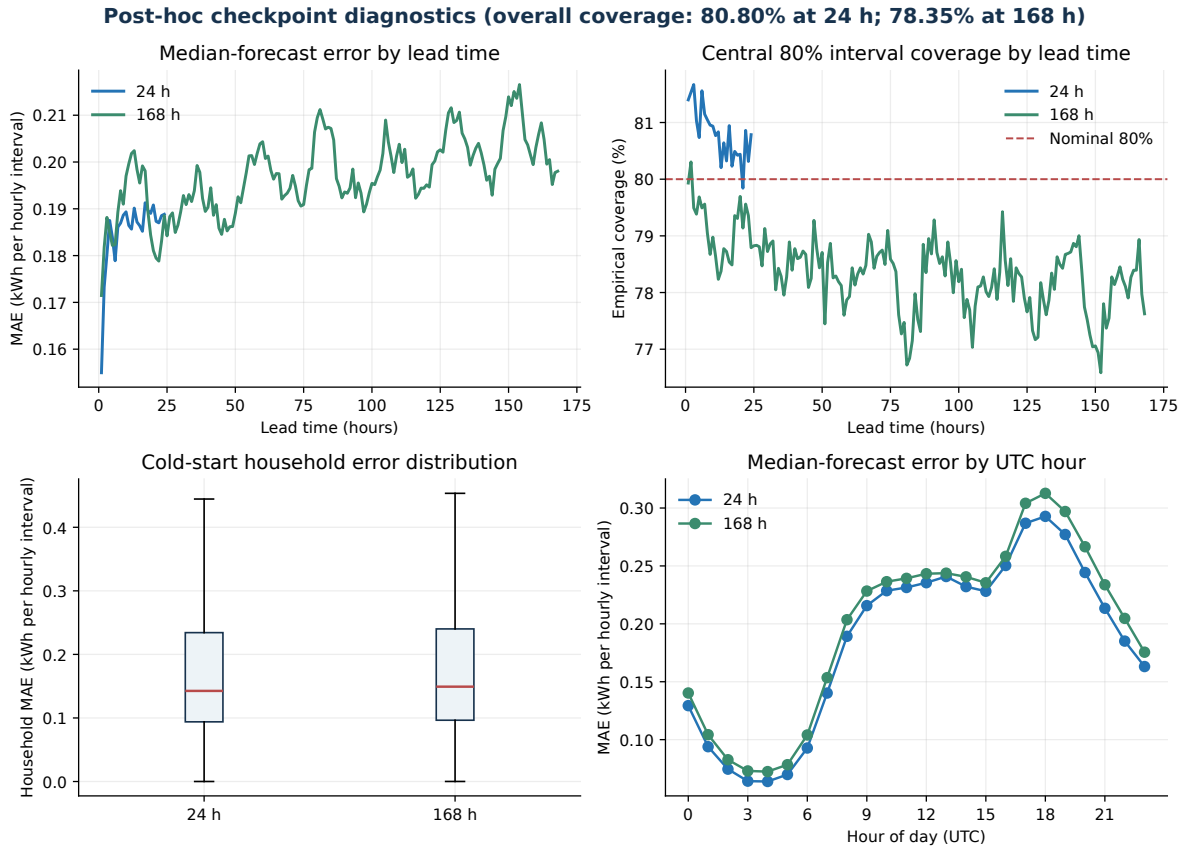


Figure 5.3 – Lead-time, household, and UTC-hour diagnostics recomputed from the frozen cold-start checkpoints. Overall metrics remain those in the original run files.

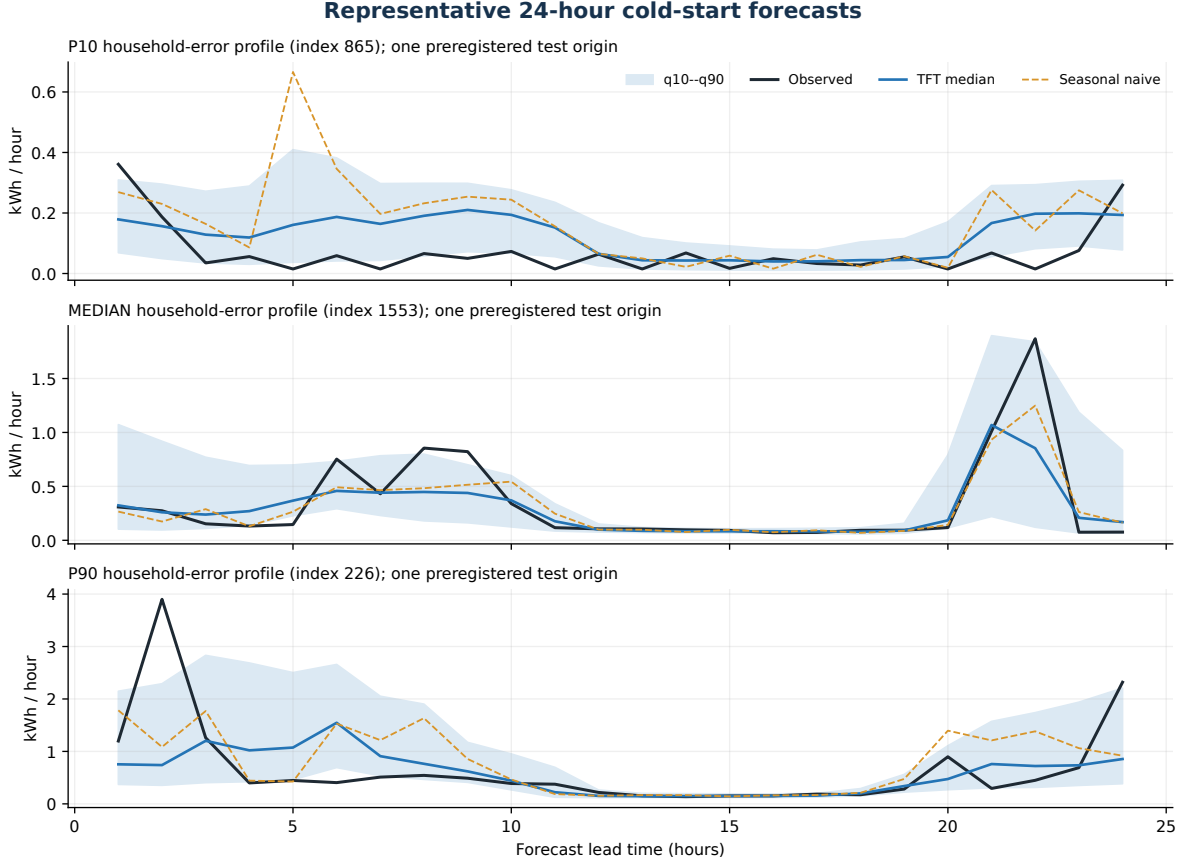


Figure 5.4 – One fixed 24-hour test origin for households nearest the p10, median, and p90 household-MAE profiles. These examples illustrate behavior and are not selected as best cases.

The diagnostic manifest records the two checkpoint hashes, raw-array and scaler hashes, script hash, Git commit, execution environment, cohort sizes, and generated CSV files under `report/evidence`. The reanalysis used an RTX 3070 Laptop GPU with PyTorch 2.6.0; it did not retrain or overwrite either checkpoint.

5.7.7 Monthly experiments

For 30 daily totals, the historical MultiCycleNet-adapted and Global N-BEATS candidates improved seasonal-naive MAE by only 8.50% and 5.31%, with macro R^2 values of -0.6078 and -0.7170. They remain archived. A subsequent production study compared zero-shot Chronos-2 and rank-8 LoRA schedules of 100, 300, and 600 updates under a frozen selection protocol. The 600-step adapter won, after which the sealed transfer diagnostics were opened.

Table 5.7 – Frozen evidence for the selected 30-day Chronos-2 LoRA release.

Evaluation	Primary result	Macro R^2	Central-80% coverage
Tetouan selection	49.82% MAE improvement over strongest frozen baseline	0.4498	88.15%
Portugal diagnostic	34.04% daily-MAE and 28.48% month-total improvement	0.4668	80.70%
Southern Morocco diagnostic	MASE 7.27% worse than baseline	-0.1769	84.44%

The release therefore passes its declared production gate without supporting a universal geographic claim. It pins the exact 120M base revision, adapter checksum, FP32 inference precision, non-negative output policy, quantile ordering, and residual-based conformal correction. A CPU benchmark measured approximately 9.04 seconds for cold loading and 76.8 ms median warm inference. Because the base is immutable and cached during the container build, ordinary startup is offline and checksum verified.

5.7.8 Same-architecture hourly challengers

The successful month architecture justified a controlled attempt at the day and week tasks. Separate rank-8 LoRA adapters were selected on known-household validation windows using 672 hours of context; the 500-household London cold cohort was opened only after the 300-step schedule won. The comparison used the exact packaged TFT inference path rather than a newly tuned proxy.

Table 5.8 – Frozen London cold-household audit of hourly Chronos-2 challengers.

Horizon	Chronos MAE	TFT MAE	MAE gain	Macro R^2	HH win rate
24 h	0.176788	0.182067	2.90%	0.2920	79.96%
168 h	0.188696	0.193770	2.62%	0.2472	80.96%

Both adapters pass the preregistered 2% MAE, non-negative R^2 , coverage, household-win, finite-output, and FP32 reload gates. Their central-80% coverages are 79.72% and 78.41%; warm CPU inference is approximately 111 ms and 127 ms. Nevertheless, activation is deferred: the improvement is modest, the required history would double from 336 to 672 hours, the first base load adds about 904 MB of resident memory, and no geographic hourly transfer gate has been run. The checksum-addressed releases remain production-eligible challengers pending a shared-base multi-adapter loader and container resource gate.

5.8 Training and inference considerations

The full-profile Global TFT training times recorded in the leaderboard are approximately 1,356.7 seconds for 24 hours and 1,666.7 seconds for 168 hours. Their optimization histories are shown in figure 5.5. N-BEATS training is shorter in the stored runs. Training time was not the primary selection metric because it is an offline cost, but checkpoint size and CPU inference compatibility matter for deployment. Both packaged TFT artifacts are approximately 5.6 MB and use the same 1,386,262-parameter architecture.

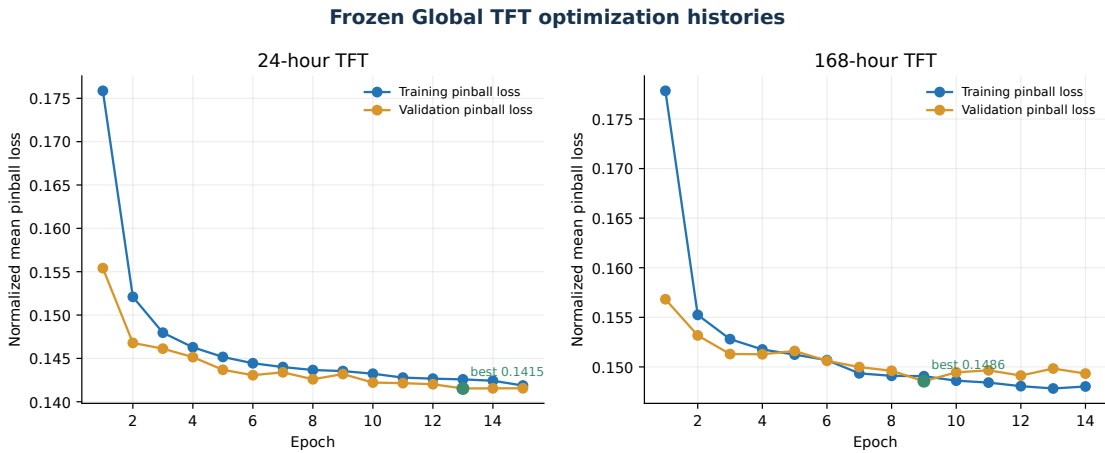


Figure 5.5 – Training and validation pinball-loss histories stored by the frozen 24-hour and 168-hour Global TFT runs.

The run files report 3.864 seconds for the complete batched 24-hour cold-start evaluation and 5.578 seconds for the complete batched 168-hour evaluation on the recorded training environment. These totals cover thousands of windows and must not be read as single-request web latency. No controlled concurrency or percentile-latency benchmark is claimed for the API.

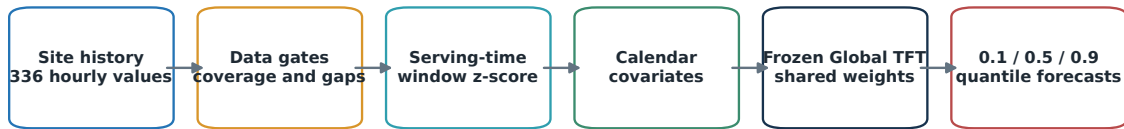
The product loads fixed artifacts and verifies their contracts. It does not retrain on user request. The active hourly paths remain the compact TFTs; the active month path uses the verified Chronos-2 adapter. This reduces latency and prevents accidental

replacement of a validated model by an untracked run.

5.9 Final selection

Global TFT remains selected for both hourly horizons because it provides strong cold-start evidence, broad household win rates, native quantiles, compact artifacts, and the most favorable current serving tradeoff. Chronos-2 LoRA is selected for the 30-day daily capability because it fills a missing horizon and passes the stronger frozen cross-dataset gate. Figure 5.6 separates the hourly operational transformation from the research evidence boundary.

Packaged Global TFT serving flow and its validation boundary



Frozen research metrics used per-household statistics fitted on each household's training segment.

The rolling 336-hour serving transform is operationally implemented but not validated by those headline metrics.

Figure 5.6 – Global TFT flow from a site's recent history to quantile forecasts.

The final selection is narrower than the research history by design. iTransformer leads the stored ECL benchmark; Global TFT remains the active hourly family; Chronos-2 LoRA is active only for 30 daily targets. The new hourly Chronos adapters are positive, reproducible challengers rather than deployed replacements. TimePro and TimeMixer++ remain attempted, historical monthly N-BEATS and MultiCycleNet remain archived, and the DEPM scores remain invalidated. Geographic transfer and per-site calibration remain scientifically conditional.

Table 5.6 – Frozen research metrics compared with supplementary serving-normalization evidence on the same cold-start samples.

Metric	Research 24 h	Serving 24 h	Research 168 h	Serving 168 h
Evaluated households	500	500	499	499
Evaluation windows	11,871	11,871	11,830	11,830
Macro MAE (kWh)	0.184928	0.181945	0.197322	0.194138
Median household MAE (kWh)	0.142566	0.142769	0.149232	0.150560
p90 household MAE (kWh)	0.362807	0.362684	0.389872	0.387758
Macro RMSE (kWh)	0.328317	0.322854	0.341090	0.335025
Macro bias (kWh)	-0.074996	-0.061638	-0.076080	-0.053130
Macro R^2	0.270362	0.285221	0.216235	0.244525
Global R^2	0.609431	0.638839	0.548724	0.581186
Seasonal-naive macro MAE (kWh)	0.251540	0.251540	0.249055	0.249055
Households beating seasonal naive	495/500	498/500	491/499	496/499
Household win rate	99.0000%	99.6000%	98.3968%	99.3988%
q10 pinball loss (kWh)	0.029654	0.029440	0.031180	0.031123
q50 pinball loss (kWh)	0.092568	0.091103	0.098576	0.096999
q90 pinball loss (kWh)	0.064440	0.061746	0.068650	0.066166
Central-80 empirical coverage	80.7963%	81.0108%	78.3518%	78.0556%
Mean interval width (kWh)	0.559770	0.571945	0.585659	0.598749
Quantile-crossing rate	0.01895%	0.00000%	0.02742%	0.00000%

EnergyAI Platform Design and Implementation

6.1 Product vision and user journey

EnergyAI turns the validated forecasting workflow into an application that a non-research user can operate without choosing neural architectures or manipulating notebooks. The primary journey is deliberately short: authenticate, configure a site and meter, import energy history or start the labelled demonstration simulator, inspect data quality, select 24 hours, 168 hours, or 30 days, generate a forecast, and review the result with its uncertainty, cadence, and provenance. Model selection is a backend responsibility governed by the artifact registry.

The platform does not expose every research feature. Product V1 prioritizes Dashboard, Usage, Forecast, Actions, and Settings. Administrative views are role-gated. Simulation is a secondary, explicitly synthetic demonstration surface; research model selection remains a backend and evidence-registry concern rather than a normal-user control.

6.2 Layered architecture

EnergyAI layered architecture

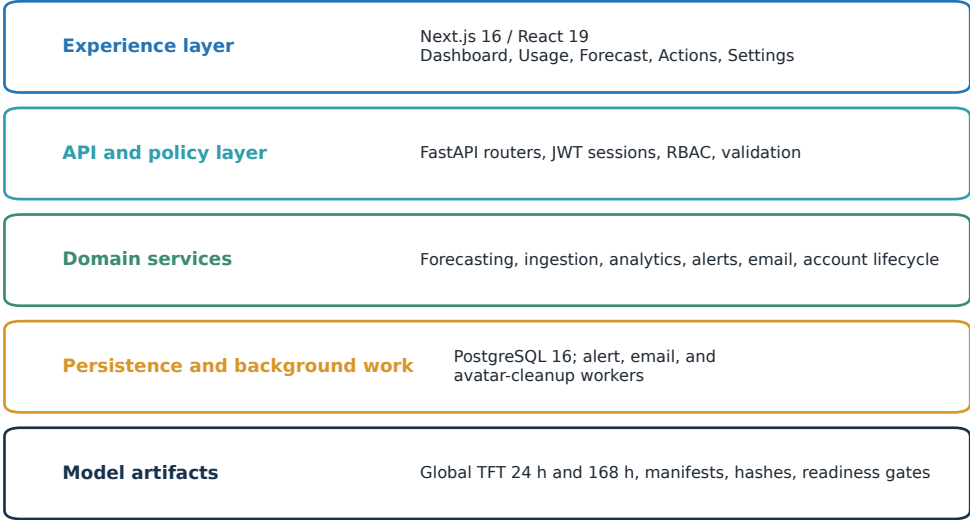


Figure 6.1 – Layered architecture derived from the current Docker Compose and application source.

The presentation layer uses Next.js 16, React 19, TypeScript, Tailwind CSS, and reusable UI components. The API layer uses FastAPI and Pydantic schemas. Persistence uses SQLAlchemy models, Alembic migrations, and PostgreSQL 16. Separate worker services handle alert, email, and avatar-cleanup responsibilities. The model layer contains fixed Global TFT artifacts plus the checksum-addressed Chronos-2 month adapter and pinned base revision.

The Docker Compose configuration defines six services: PostgreSQL, backend, frontend, alert worker, email worker, and avatar-cleanup worker. At the verified local checkpoint, database, backend, and frontend health checks pass and the three workers run. The public Azure topology deploys the web, API, and email-worker containers, with PostgreSQL provided as a managed service. Alert and avatar-cleanup workers remain local-only. This distinction prevents local worker readiness from being misreported as cloud worker availability.

6.3 Frontend design

The Next.js application provides authenticated and public routes. Public routes include the landing page, registration and login, password recovery, email verification states, privacy, terms, and support. The normal-user navigation contains exactly five primary

destinations: `/dashboard`, `/usage`, `/forecast`, `/actions`, and `/settings`. Compatibility routes redirect `/consumption` to Usage; `/alerts` and `/recommendations` to Actions; and `/reports` to Usage. The `/admin` view is rendered only for an administrator.

The interface uses explicit empty, loading, blocked, error, and success states. This is particularly important for forecasting. A missing model, insufficient history, disabled capability, or failed data-quality gate must be explained rather than replaced by a synthetic chart. Horizon state is persisted across 24-hour, 168-hour, and 30-day views. Daily labels, cards, chart axes, peak descriptions, and PDF names are derived from the returned cadence instead of dividing an assumed hourly series. A new empty-meter simulator bootstrap supplies a complete year of hourly context, so all three model contracts become usable without lowering the 270-day month gate or padding the TFT inputs.

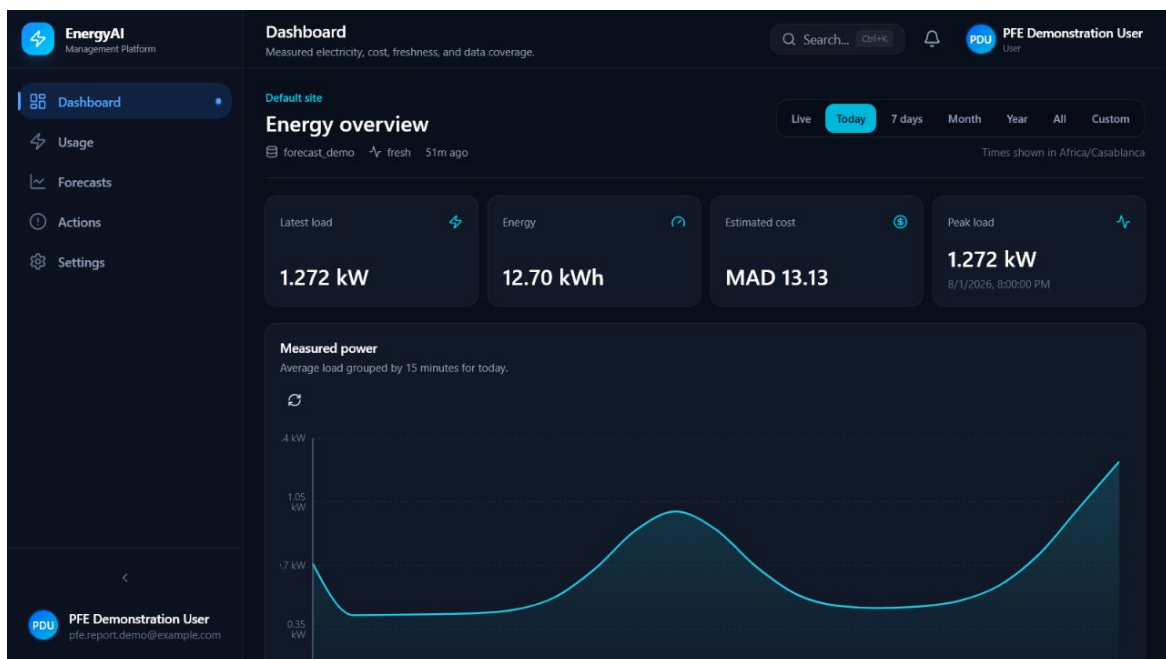


Figure 6.2 – Normal-user EnergyAI dashboard with measured hourly consumption, tariff-cost estimates, source freshness, and data coverage. The non-personal demonstration account has no administrative navigation.

6.4 Backend API and domain model

FastAPI routers separate authentication, account lifecycle, settings, ingestion, consumption, forecasting, analytics, alerts, recommendations, simulation, monitoring, system health, and administration. Importing the current application and generating its OpenAPI schema yields exactly 80 method–path operations across 74 unique paths, including the two root health operations. Pydantic request and response models define validation boundaries, while dependency functions enforce authentication, role, and ownership

rules.

The core domain model is shown in figure 6.3. A User owns one or more Sites; a Site contains Meters; meter observations are stored as SmartMeterReading records; IngestionBatch records summarize import operations; Forecast records preserve output and horizon provenance; AlertConfig and Alert represent policy and detected conditions; Recommendations record suggested actions. Additional entities support refresh tokens, account-action tokens, OAuth challenges, outbox delivery, audit events, budgets, and system settings.

Core domain class diagram

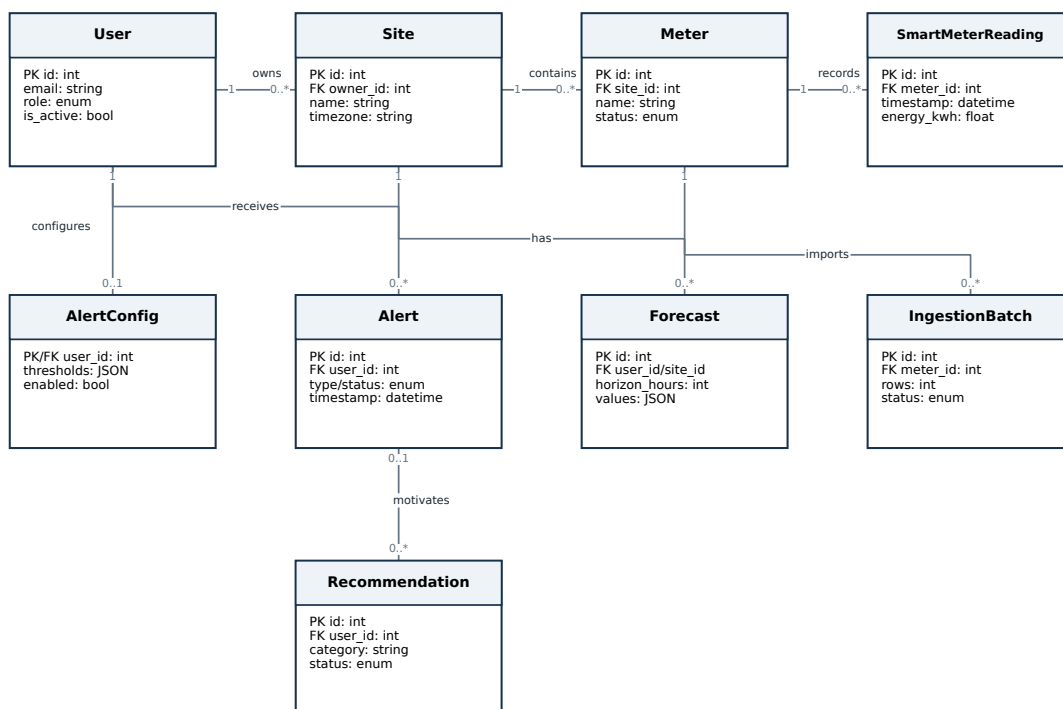


Figure 6.3 – Core EnergyAI domain class diagram based on the SQLAlchemy entities.

6.5 Authentication, JWT sessions, and RBAC

Local login verifies a bcrypt password hash. On success, the backend returns an access token and sets refresh state through an HttpOnly cookie. The server stores refresh-token hashes and supports rotation, logout, and logout-all operations. This design reduces exposure compared with storing long-lived refresh credentials in browser local storage.

Role-Based Access Control (RBAC) protects administrative operations. Ownership checks are also required for normal-user resources: role membership alone must not permit access to another user's site, readings, forecast history, or alerts. Account actions cover email verification, password reset, Google identity linking where configured, data

export, avatar management, and irreversible deletion with recent-authentication controls. Permanent deletion performs explicit ordered, set-based deletes across audit history, outbox state, OAuth challenges, recommendations, alerts, forecasts, budgets, readings, ingestion batches, meters, simulator sessions, sites, tokens, linked identities, and the user. This avoids loading thousands of readings and invoking an unbounded ORM cascade inside one request. The transaction preserves only a new anonymized `account.deleted` audit event after the ownership graph is removed.

Sequence diagram - local authentication

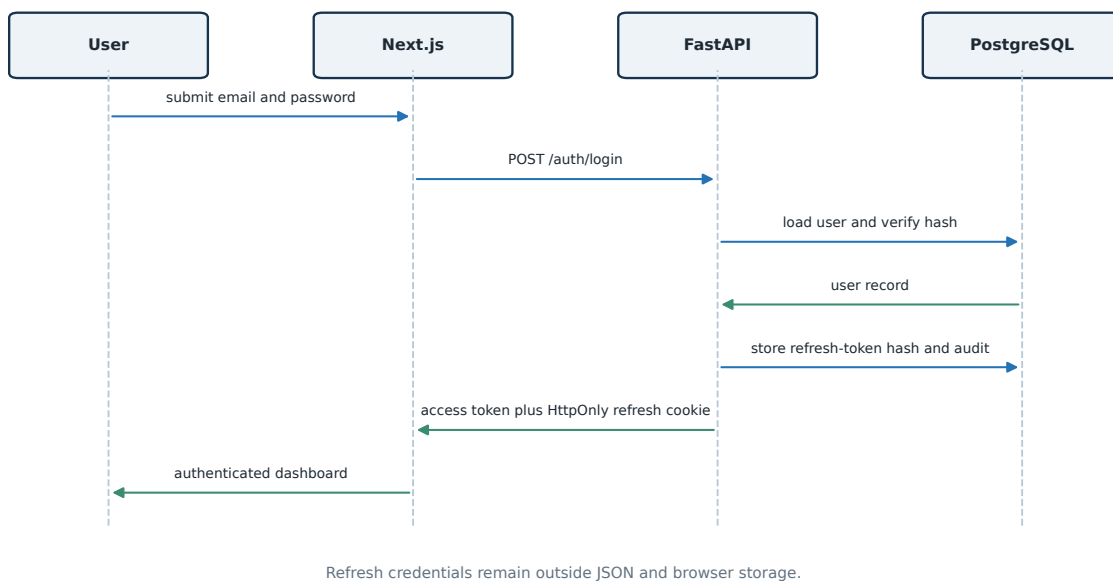


Figure 6.4 – Sequence diagram for local authentication and refresh-state creation.

6.6 CSV ingestion and validation

The product input contract is intentionally simple: timestamps and energy values in kWh. A meter-scoped preview endpoint parses a bounded sample and returns detected columns and validation information. Import is a separate confirmed operation. The backend verifies ownership, normalizes supported timestamps, rejects invalid rows, bounds file/row processing, stores accepted readings transactionally, and records an ingestion batch.

This separation improves safety and usability. A user can identify a wrong delimiter, timestamp column, or unit before modifying persisted history. Duplicate or malformed timestamps are not silently converted into model-ready data. CSV preview remains available while the simulator runs. At confirmed import time, the client refreshes simulator status and warns that continuing the live synthetic feed could mix sources. Consent stops the simulator before import; cancellation writes nothing. Historical

simulation rows remain labelled and are not automatically deleted. The simulator page provides a matching “Stop demo and use real data” action that links directly to the importer. The consumption API supports period summaries, paginated readings, current/history statistics, and export.

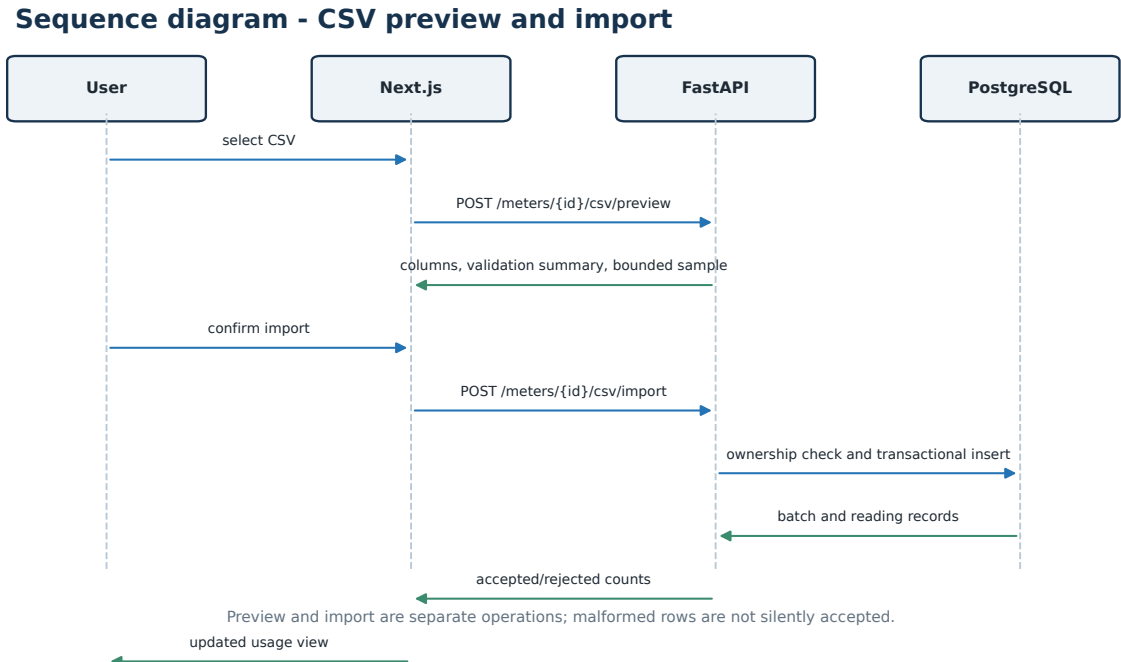


Figure 6.5 – Sequence diagram for CSV preview and confirmed import.

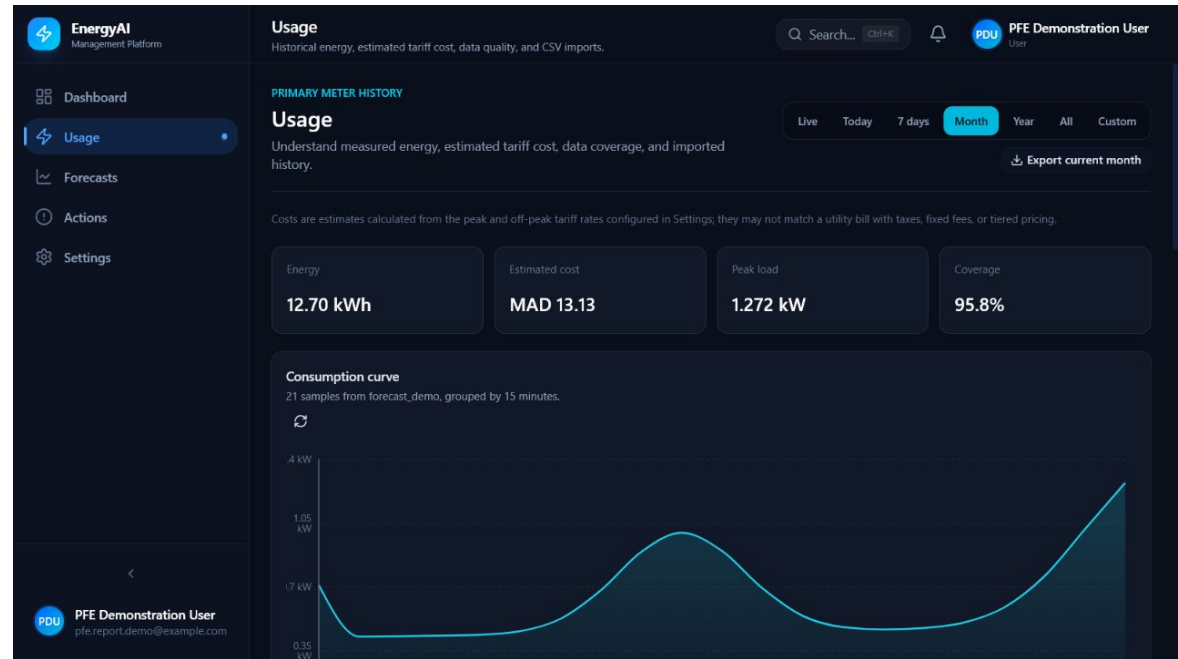


Figure 6.6 – Normal-user Usage view showing persisted demonstration readings, monthly energy and cost summaries, provenance, and data coverage.

6.7 Persistent deterministic demonstration history

The demonstration simulator is persistent and source-aware. On its first start for an empty primary meter, it creates 365 days of household history at hourly intervals, or 8,761 timestamped readings including both boundary points. Hourly cadence is deliberate: it is the native input resolution of the TFTs and can be summed directly into the daily Chronos context, while requiring about one quarter of the rows of a year-long 15-minute bootstrap. Each value is a deterministic function of a site-specific seed, timestamp, and profile version. The profile combines overnight base load, morning and evening peaks, weekday and weekend differences, and bounded variation. Every generated row uses `source="simulation"` and simulated quality metadata.

Azure may scale the API to zero while the simulator session remains logically running. On the next backend start, catch-up advances only from the latest eligible simulation reading and never searches for arbitrary historical holes. A continuity marker prevents the feature from rewriting gaps that existed before deployment; stopping and later restarting the simulator establishes a new boundary so the stopped interval also remains visible. Ordinary catch-up retains the 15-minute cadence. Exceptionally long outages select progressively coarser intervals and never exceed 5,000 generated points. The unique meter/timestamp constraint and deterministic ingestion keys make bootstrap and catch-up idempotent under replay.

The explicit “Reset demo data” operation deletes and regenerates only simulator-owned readings and batches. CSV and push readings remain unchanged, login never generates history, and a non-empty imported or measured meter is not automatically bootstrapped. This boundary provides 8,760 hours of reproducible context, satisfying the 336-hour TFT and 365-day Chronos lookbacks without changing either model’s validated input contract.

6.8 Forecasting service and artifact contracts

The forecast router exposes capabilities, readiness, demo-history preparation, run, latest, and history operations. For 24 and 168 hours, the service validates and normalizes the latest 336-hour window before TFT inference. For the month capability, it integrates persisted meter intervals into rolling 24-hour energy blocks, requires at least 270 complete daily values within a maximum 365-day context, performs causal interpolation only across bounded internal missing days, and invokes the checksum-verified Chronos-2 LoRA release. Forecast records preserve target count, interval length, resolution, input range, and model provenance so that retrieval and PDF generation cannot reinterpret 30 daily targets as 720 hourly points.

Forecast request lifecycle

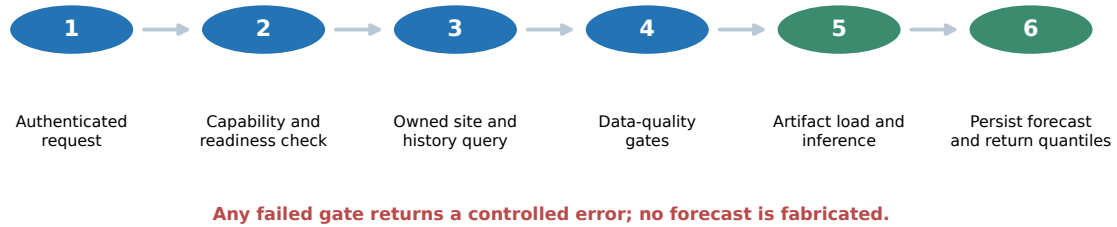


Figure 6.7 – Forecast lifecycle and failure boundaries.

The two TFT manifests record artifact version, checksum, size, unit, lookback, horizon, normalization, quantiles, calendar features, data gates, research metrics, and limitations. The month manifest additionally pins the Chronos-2 base revision and checksum, LoRA checksum, FP32 inference, 365/270-day context contract, 30-target daily output, conformal correction, transfer diagnostics, and CPU benchmark. The container build installs the pinned Chronos/PEFT runtime and prefetches the exact base weights; runtime loading is offline. Artifact readiness is fixed and explicit rather than inferred from a mutable database label.

Sequence diagram - 24-hour or 168-hour forecast

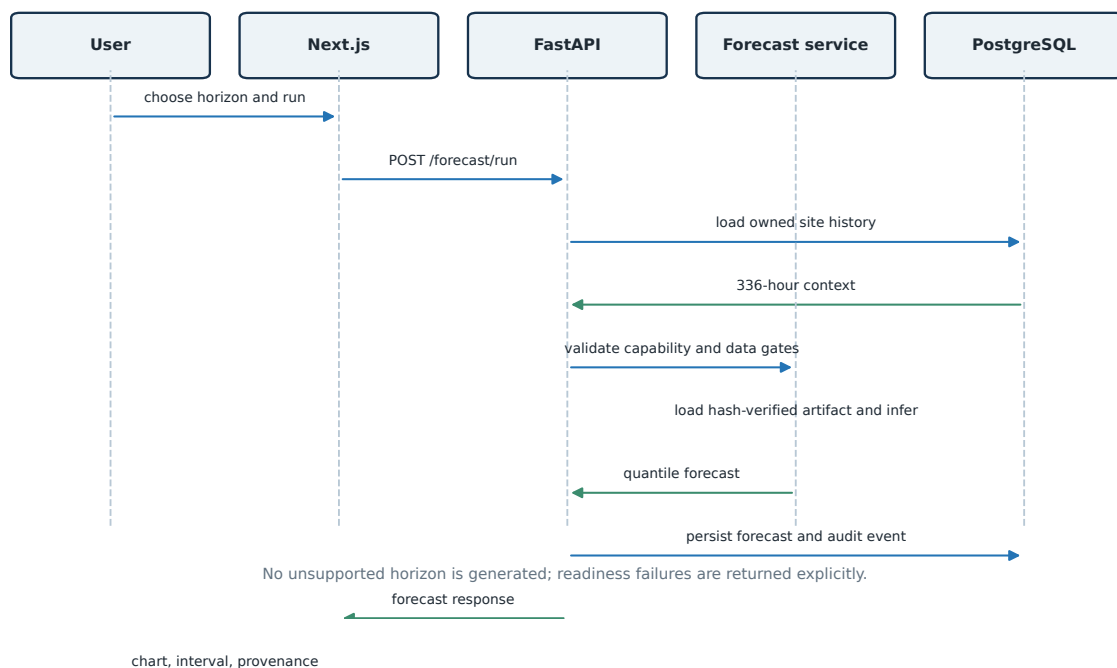


Figure 6.8 – Sequence diagram for a supported forecast; the month path uses daily aggregation and a separate foundation-model contract.



Figure 6.9 – Normal-user 24-hour Global TFT forecast with readiness gates, hourly quantiles, and explicit research-evidence context.

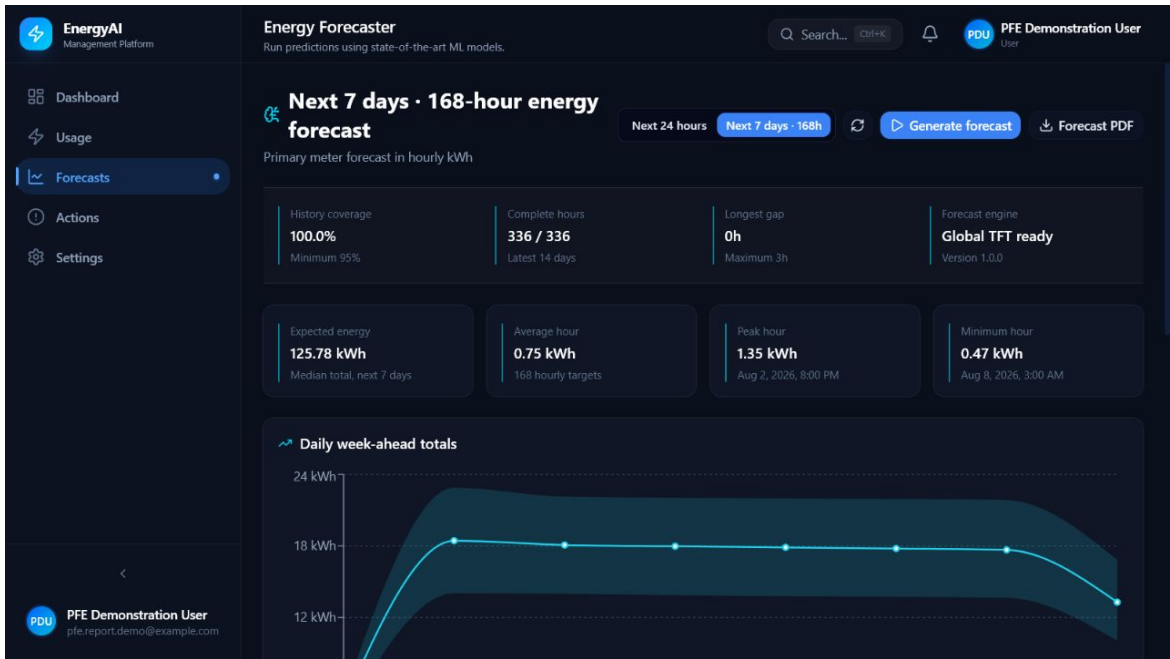


Figure 6.10 – Normal-user 168-hour Global TFT forecast with the same input-readiness contract and daily aggregation of 168 hourly targets.

6.9 Dashboard, actions, alerts, and reports

The dashboard aggregates current consumption, historical trend, forecast summary, and prioritized actions from database-backed APIs. Its “At a glance” narrative explains

the selected period before requiring a user to decode the cards: how much energy and estimated cost were observed, when the peak occurred, how the period compares, whether data are fresh and sufficiently covered, which sources contributed, and what action is most relevant. Simulation provenance receives a prominent trust message whenever synthetic readings contribute. The Usage area focuses on measurements and export. Forecast focuses on horizon selection, readiness, quantiles, validation context, and history. Actions combine recommendations and alert state so that the prediction can lead to a practical response rather than remain an isolated chart.

Alerts have acknowledged, resolved, and reopened states and a per-user configuration. Background workers support alert generation and delivery responsibilities. Recommendations can also be updated through explicit status transitions. PDF export is available from the forecasting workflow through the analytics endpoint. The legacy `/reports` page redirects to Usage; there is no separate reports workspace or external reporting warehouse.

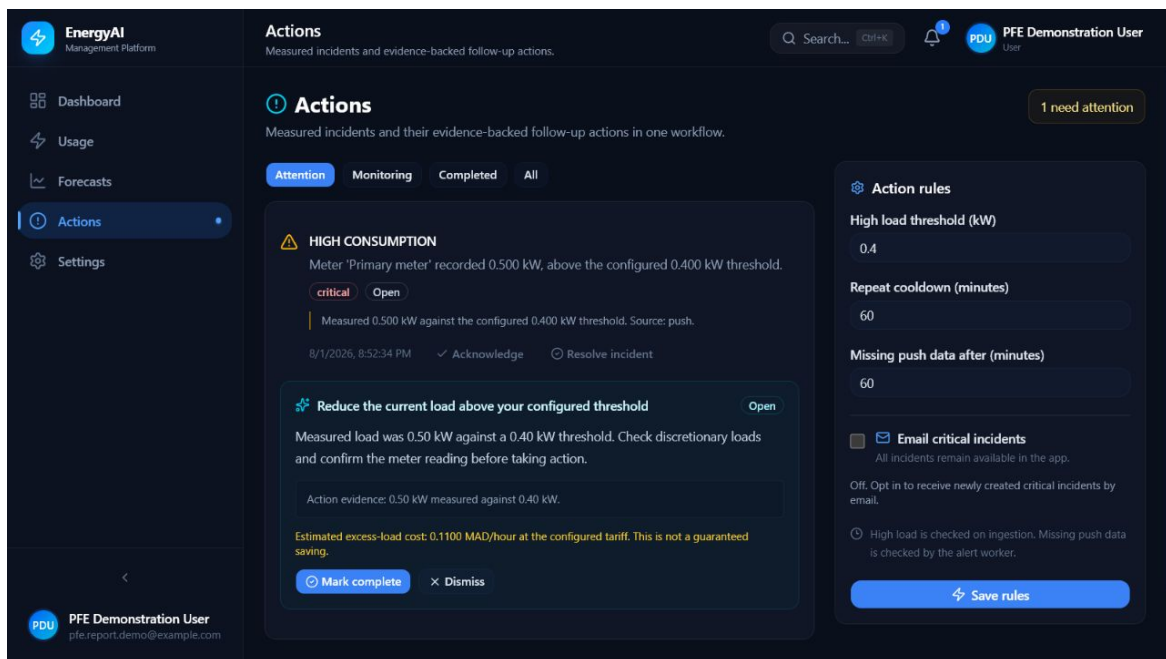


Figure 6.11 – Normal-user Actions view after a measured 0.500kW push sample crossed the configured 0.400 kW threshold, producing an evidence-backed incident and follow-up action.

6.10 Real-time behavior and background work

The repository contains WebSocket connection management and real-time alert behavior, along with workers for alerts, email outbox delivery, avatar cleanup, and simulator continuity. Their responsibilities are separated from ordinary user reads. Simulator history is created only by explicit simulator lifecycle operations or the startup catch-up of a session that remained running; login and dashboard requests do not

write readings. The Azure deployment runs the email worker with the existing SMTP configuration, and the API reports email delivery ready. The live Google Web client also opens the account chooser for the authorized Azure origin. Provider availability remains operationally dependent, while alert and avatar-cleanup workers are not deployed in Azure.

6.11 Observability and error handling

System endpoints expose liveness, readiness, health, version, and legal information. Administrative endpoints expose user, system, and model-readiness views. Structured logging and audit events preserve significant account, authentication, ingestion, and forecast actions. User-facing errors use explicit codes and messages rather than leaking internal exception details.

The product must distinguish service readiness from model readiness. A healthy FastAPI process can still lack a valid forecasting artifact or sufficient site history. Conversely, a valid artifact does not make external email or OAuth providers ready. Separate capability responses allow the frontend to present these states accurately.

6.12 UI/UX and accessibility considerations

The interface uses a consistent dark visual language, responsive layouts, labeled controls, status messages, concise narrative interpretation, and chart alternatives where practical. Browser tests cover desktop Chromium, Firefox, WebKit, and 360, 390, and 768-pixel responsive projects. Accessibility is not claimed as certified: a formal WCAG audit, keyboard-only review, screen-reader validation, and contrast inventory remain future work.

6.13 Local Docker deployment

The verified local deployment profile is reproducible:

```
docker compose up --build -d
docker compose ps
curl http://localhost:8000/api/v1/system/live
curl http://localhost:8000/api/v1/system/ready
```

Environment configuration must replace insecure sentinel secrets before startup. PostgreSQL data and avatars use named volumes. Alembic migrations run against the configured database. Model artifacts are packaged in the backend image or mounted according to the repository's Compose build context. The production build includes

the exact month adapter and cached base model; deployments may disable the month capability explicitly but do not download mutable weights at request time.

6.14 Azure platform deployment

The Product V1 jury release was updated in Microsoft Azure’s Italy North region on 15 August 2026. Web, API, and email-worker revision `v4-d25295b` integrates the 365-day simulator bootstrap, the 24-hour and 168-hour TFTs, and the 30-day daily Chronos-2 capability. The verified `origin/main` source commit and immutable image digests are recorded in the deployment evidence registry. The public endpoints are:

EnergyAI web application

`https://ca-energyai-web.calmtree-b020e00c.italynorth.azurecontainerapps.io`

EnergyAI API

`https://ca-energyai-api.calmtree-b020e00c.italynorth.azurecontainerapps.io`

Verified Product V1 deployment on Microsoft Azure

Resource group `rg-energyai-pfe` - Italy North - verified 9 August 2026

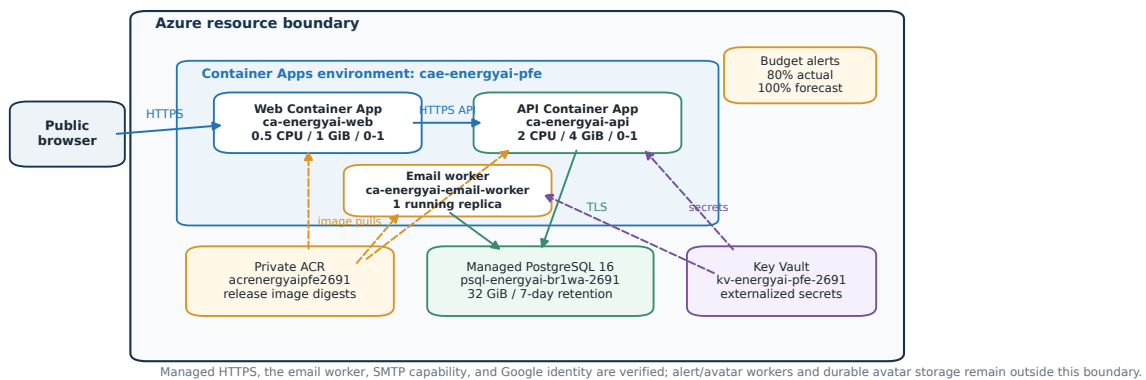


Figure 6.12 – Verified Azure deployment topology for Product V1. Solid arrows show runtime traffic; dashed arrows show image or secret supply.

Table 6.1 – Verified Azure deployment boundary on 15 August 2026.

Layer	Azure resource	Verified state
Frontend	<code>ca-energyai-web</code>	Revision <code>v4-d25295b</code> ; external HTTPS ingress; zero-to-one replicas; healthy with 100% traffic
API and models	<code>ca-energyai-api</code>	Revision <code>v4-d25295b</code> ; external HTTPS ingress; two vCPU and 4 GiB; healthy with 100% traffic; 24 h, 168 h, and 30-day artifacts warmed
Email worker	<code>ca - energyai - email - worker</code>	Revision <code>v4-d25295b</code> ; one healthy replica; PostgreSQL outbox and SMTP configuration preserved
Persistence	<code>psql - energyai - br1wa - 2691</code>	PostgreSQL 16, Burstable B1ms, 32 GiB; seven-day backup retention; application database ready
Image supply	<code>acrenergypfe2691</code>	Private Basic registry; images pinned to release-specific SHA-256 digests
Secrets	<code>kv-energyai-pfe-2691</code>	Database, JWT, and administrator secrets generated and retained outside the repository
Cost control	Subscription budget	Monthly threshold of 25 in the subscription billing currency; email notifications at 80% actual and 100% forecast spend

The release smoke test returned HTTP 200 for the frontend, Login, Register, Privacy, and Terms pages and `ready=true` for the API. The readiness payload reported PostgreSQL ready and the 24-hour, 168-hour, and 720-hour artifacts available, enabled, and warmed; the 720-hour capability identifies the Chronos-2 LoRA model and returns 30 daily targets. The live authentication capability response reported email and Google enabled. Unauthenticated canonical and trailing-slash alert requests both returned HTTP 401 without a downgrade redirect. Azure revision checks confirmed the web, API,

and email-worker revisions above healthy, with one ready API and worker replica. The full browser matrix and model-enabled Compose stack also passed on the remote CI runner before promotion.

This is a controlled public PFE deployment rather than a claim of complete production maturity. The deployment uses Azure-generated hostnames rather than a custom domain. Public email registration, verification delivery, the email worker, and Google identity are enabled, but no sustained provider/load test is claimed. Alert and avatar-cleanup workers from Compose are not deployed, uploaded avatars are not durably persisted, centralized Log Analytics retention is disabled, the registry uses deployment credentials rather than workload identity, and no backup restoration, load, disaster-recovery, formal accessibility, or penetration test has been completed. The PostgreSQL firewall accepts Azure-originated services but no temporary developer-IP rule remains. The monthly budget sends notifications but is not an automatic application shutdown policy. Older revisions remain inactive; rollback selection must preserve compatible web and API versions.

6.15 Implementation summary

EnergyAI's engineering contribution is an inspectable boundary between scientific artifacts and product behavior. The application exposes only supported capabilities, distinguishes hourly and daily cadence, verifies artifact identity, rejects insufficient input, records provenance, isolates user data, preserves uncertainty, labels every synthetic reading, and keeps simulator continuity separate from measured-data gaps. The active release deliberately keeps the compact TFTs for day/week and uses Chronos-2 only where it fills the missing month capability. The next chapter evaluates the quality evidence for this implementation and states what remains conditional.

Validation, Quality, Discussion, and Limitations

7.1 Validation scope and evidence date

The software-validation evidence in this chapter begins with the full application audit performed on 31 July 2026 and includes the foundation-model work completed on 14 August and the coordinated Azure promotion completed on 15 August 2026. Historical reports are retained for chronology because test counts, routes, and capabilities changed over time. The audit uses backend tests, frontend quality commands, browser tests, artifact checks, real-wrapper smoke tests, source review, immutable image digests, and public cloud readiness. GitHub Actions run `31893388429` verifies source commit `d25295b`; Azure revisions with the same suffix serve the verified web, API, and email-worker release.

7.2 Backend validation

The current local suite passed 116 tests with four expected PostgreSQL-only skips under SQLite. The new cases cover the month manifest and checksum, 365-day input preparation, 30-target inference, cadence-aware API persistence and reload, monthly PDF output, and the hourly challenger protocol. A focused real-wrapper smoke test also loaded the exact Chronos-2 base and adapter and returned 30 finite ordered quantiles. In the earlier full audit, a uniquely named disposable PostgreSQL database was migrated through all 17 Alembic revisions, tested, and removed; that historical result remains 100 passed and zero skipped. No schema migration was required because the existing integer horizon and JSON forecast snapshot can persist the new cadence metadata.

The PostgreSQL run is the stronger result because it exercises locking and database behavior that SQLite cannot reproduce. It also confirms that migrations can construct the schema from empty state. The audit did not mutate the live product database.

No coverage percentage is claimed. Test count and coverage are different measures; passing suites do not prove that every path is exercised.

7.3 Frontend and browser validation

ESLint passed, TypeScript type checking passed, and the production build completed with 25 routes. The current Playwright matrix executes 198 tests across desktop Chromium, Firefox, WebKit, and three responsive projects. It covers 360-pixel, 390-pixel iPhone, and 768-pixel profiles. Simulator coverage verifies the one-year hourly bootstrap, explicit synthetic provenance, all-model readiness message, reset isolation, and handoff to real CSV data. Populated 167-hour and 169-hour local-week tests continue to exercise daylight-saving boundaries.

Table 7.1 – Current software validation summary.

Area	Result	Interpretation
Backend local suite	116 passed, 4 skipped	Current local suite; PostgreSQL-only behavior excluded
Backend CI with PostgreSQL	Passed	Empty database migrated; suite passed locally and in test image
Backend on isolated PostgreSQL	100 passed, 0 skipped	Historical full-audit evidence retained for chronology
Frontend lint and type checking	Passed	Static quality gates are green
Frontend production build	Passed, 25 routes	Current route tree compiles for production
Playwright six-project matrix	198 passed, 0 failed	Desktop and three responsive profiles green
Docker Compose	Model-enabled core health checks green	Remote production stack validated; public status evaluated separately below
Azure public deployment	Web/API/worker v4-d25295b; healthy	Managed PostgreSQL and all three model horizons warmed; alert/avatar workers remain absent

7.4 Model-contract and artifact validation

The product does not rely only on a database model name. Artifact tests verify manifest parsing, supported capability, checksum, strict load, CPU inference, finite output, expected shape, cadence, and reference points where recorded. The two active TFT checkpoints share a file size of 5,600,105 bytes but have distinct SHA-256 hashes.

The month release separately verifies a 4,854,584-byte LoRA adapter, the pinned Chronos-2 base revision, FP32 inference, and exactly 30 ordered daily quantiles. A real-wrapper smoke test completed a checksum-verified load and prediction.

The manifests also preserve limitations. London-cohort metrics are not a guarantee for a client site. New hourly requests use their recent 336-hour window for normalization, while month requests require a substantially longer daily history. The month model’s positive Tetouan and Portugal evidence coexists with a negative southern-Morocco diagnostic, and its conformal correction is not a per-site guarantee. The hourly Chronos adapters pass their London gates but remain inactive because deployment value includes memory, startup, input migration, and geographic evidence as well as MAE.

7.5 Security validation

The audit reviewed password hashing, JWT and refresh-token behavior, account-action tokens, Google identity gating, role checks, ownership filters, upload bounds, request validation, SSRF controls, and secret configuration. CI includes a full-history gitleaks job. No P0 authentication bypass or data-loss defect was identified. Permanent account deletion is additionally protected by recent password reauthentication and a literal confirmation value. Its implementation removes the complete ownership graph with ordered set-based statements, then writes only an anonymized deletion audit event, avoiding the request timeout observed with the earlier row-by-row ORM cascade.

One current security-quality concern remains: the frontend application does not define its own complete security-header policy such as CSP, frame policy, content-type options, or referrer policy. Azure supplies managed HTTPS ingress, but transport security does not by itself establish a complete application-header policy. After dependency overrides and lockfile normalization, the final npm audit and the production Python requirements audit reported zero known vulnerabilities. Google integration uses only the public client ID; no provider secret is required or stored by EnergyAI.

7.6 Docker and deployed service health

The six-service Compose stack ran with PostgreSQL, backend, and frontend healthy and all workers running. For the new release, Compose configuration resolves with the month feature flag and the production Dockerfile stages the exact adapter, pinned dependencies, and base-model prefetch step. GitHub Actions built and started the full foundation image, and the same immutable backend digest passed Azure startup and warm-up at two vCPU and 4 GiB.

The final Azure smoke test confirmed public TLS and HTTP 200 for the principal pub-

lic pages, API liveness and readiness, managed-database connectivity, warmed 24-hour, 168-hour, and 720-hour models, ready SMTP capability, and a healthy one-replica email worker. Google identity remains published for external users; the earlier account-chooser test created no user. A read-only Azure check found `ca-energyai-web-v4-d25295b`, `ca-energyai-api-v4-d25295b`, and `ca-energyai-email-worker-v4-d25295b` healthy, with 100% traffic on the web and API release. The web and API retain zero-to-one scaling; the email worker remains at one replica. Operational readiness is therefore demonstrated for a controlled public PFE presentation, while production hardening remains conditional on security headers, durable avatar storage, centralized monitoring, Azure alert/avatar workers, restore evidence, load testing, and disaster-recovery procedures.

7.7 Requirement-to-test traceability

The report’s traceability registry maps each major requirement to code and validation evidence. Examples include JWT authentication to authentication tests, RBAC to admin authorization tests, per-user data isolation to PostgreSQL integration tests, CSV pre-view/import and simulator handoff to ingestion/browser tests, persistent demonstration continuity to simulator tests, bounded account deletion to ownership-graph tests, and each forecast horizon to model-contract and dual-horizon browser tests. The complete machine-readable matrix is stored in `report/evidence/requirements_traceability.csv`.

Traceability does not remove the need for judgment. For example, a route and test can show that account deletion works, but legal compliance requires institutional and jurisdiction-specific review. Likewise, an artifact test proves file identity and output shape, not real-world forecast calibration.

7.8 Scientific limitations

7.8.1 Geographic transfer

The final model evidence comes from Low Carbon London. Climate, building characteristics, tariffs, appliance ownership, and daily routines can differ materially in Morocco. Cold-start evaluation within London is stronger than random household windows, but it is not cross-country validation. A Moroccan pilot cohort is required before making local performance guarantees.

7.8.2 Input scope

The product contract uses timestamp and energy. This lowers onboarding friction but excludes weather, occupancy, tariff events, and building metadata that may explain

regime changes. Calendar features are deterministic, not measurements. Future work should compare the marginal value of weather forecasts and static site attributes under a strictly causal protocol.

7.8.3 Uncertainty calibration

The TFT produces 0.1, 0.5, and 0.9 quantiles, but the manifests state that they are not recalibrated for each client. Reanalysis under the original research scaler finds 80.7963% empirical coverage for the nominal central 80% interval at 24 hours and 78.3518% at 168 hours, with mean widths of 0.559770 and 0.585659 kWh. Under serving normalization, coverage is 81.0108% and 78.0556%, with widths of 0.571945 and 0.598749 kWh. This closes the scaler-specific comparison on the frozen London cohort, but not site-level calibration. Conformal time-series methods are a candidate extension if their assumptions, calibration cohort, and deployment contract are stated explicitly [21].

7.8.4 Research-to-serving preprocessing

The cold-start experiment gives every held-out London household a mean and standard deviation fitted on its own historical training segment; the application instead estimates both statistics from the current 336-hour request window. The preserved artifacts were sufficient to rerun exactly the same cold households, fixed origins, histories, targets, seasonal baselines, calendar features, and checkpoints with the production rolling normalization and output transform. Macro MAE is 0.181945 kWh at 24 hours and 0.194138 kWh at 168 hours under that serving transform, compared with the unchanged research headlines of 0.184928 and 0.197322 kWh. This resolves the identified transform mismatch for the frozen LCL cohort. It does not establish expected accuracy for an arbitrary new site because geography, data quality, time-zone handling, and calibration may differ.

7.8.5 Monthly horizon

The earlier monthly N-BEATS and MultiCycleNet candidates remain invalid for promotion because their cold-start macro R^2 values are negative. The replacement Chronos-2 release passes its frozen selection gate with 49.82% improvement and positive macro R^2 , then transfers positively to Portugal. Its southern-Morocco diagnostic is worse than the comparator and therefore prevents a broad geographic claim. The repository capability is production-eligible for controlled use, not universally validated.

7.8.6 Experiment completeness

TimePro and TimeMixer++ were attempted but lack completed leaderboard evidence. TSMixer appears in an exploratory weather notebook without a frozen successful metric artifact. MultiCycleNet was adapted locally and requires final primary-reference verification before any claim of exact reproduction. These gaps are reported rather than filled with inferred results.

7.9 Product limitations

The release has verified public web and API URLs, email registration and delivery capability, a healthy email worker, and a production Google identity origin, but that does not make every repository capability operational in the cloud. Alert and avatar-cleanup workers are absent; avatars are ephemeral; centralized log retention is not configured; and backup restoration has not been tested. The Azure-generated hostname is not a custom domain, the registry is not yet integrated through workload identity, and the budget is an alerting control rather than an automatic shutdown. Application security headers still require review. Formal accessibility, load, disaster-recovery, sustained provider, and multi-tenant penetration testing remain incomplete. The research tree also contains historical assets that should remain separated from the deployable product snapshot.

7.10 Threats to validity

Table 7.2 – Threats to validity and mitigation.

Validity type	Threat	Mitigation and remaining gap
Internal	Leakage, preprocessing scope, repeated test use	Causal windows, chronological split, train-only transforms, evidence registries; historic notebooks remain separately labelled
Research-serve	Research scaler differs from 336-hour serving scaler	Exact frozen-cohort comparison completed; new-site transfer and calibration remain required
Construct	Accuracy/classification confused with forecasting value	Continuous target, physical-unit errors, seasonal baselines, household distribution
External	London/IHEPC results transferred to Moroccan users	Cold-start household split; local-country evaluation still required
Conclusion	Single seed, clock-influenced dynamic sampling, aggregation choice	Frozen seed and common protocol, macro/median/p90/win rate; bitwise retraining and multi-seed finalist study remain desirable
Implementation	Third-party repository/version differences	Isolated workers and source notes; incomplete models excluded from rankings
Operational	Public endpoint interpreted as complete production maturity	Exact Azure boundary recorded; absent workers, ephemeral storage, monitoring, and recovery tests remain explicit

7.11 Discussion

The project demonstrates a change in what counts as success. At the beginning, success meant matching a 98% article result. After the audit, success meant constructing

a task that could fail honestly. That shift led to lower and more interpretable metrics, stronger baselines, broader household analysis, and explicit artifact limitations.

The final model decisions are credible because they combine causal targets, frozen protocols, strong comparators, distributional metrics, transfer diagnostics, checksummed artifacts, and explicit operational gates. The application adds strict loading and refuses unsupported cadence or history. The decision not to activate the slightly more accurate hourly Chronos challengers is part of this rigor: a production model must improve the whole system contract, not only one metric. These facts support a controlled demonstration and a defensible PFE contribution, but they do not establish universal geographic transfer or site-specific calibration.

The remaining conditional findings also improve rather than weaken the report. They separate verified public availability from full production maturity and prevent a controlled academic deployment from being misrepresented as a globally validated service.

General Conclusion

This PFE addressed a question that is both scientific and practical: how can an electricity-consumption forecasting system be made reliable when initially impressive performance is affected by target leakage and unrealistic evaluation? The work began with the replication of a hybrid DEPM architecture and reproduced classification scores close to those reported in the starting article. A critical audit then showed that direct target information, the physical relationship between active power, voltage, and intensity, non-shifted rolling features, preprocessing before splitting, and random temporal separation could explain the apparent performance. A simple logistic-regression diagnostic reached 99.4% under the suspect formulation, demonstrating that architectural complexity was not the source of the score.

The main scientific contribution was the reformulation of the task. Energy consumption became a continuous future trajectory; windows were causal; train, validation, and test regions were chronological; transformations were fitted on training observations; predictions were evaluated in physical units; and seasonal baselines were mandatory. The research compared recurrent, convolutional, patch-based, inverted-Transformer, decomposition, and global-model families while preserving the distinction between completed, incomplete, historical, and production experiments.

The ECL benchmark found iTransformer to have the lowest normalized MAE among the four completed architectures at all stored horizons. Product selection, however, used deployment-matched tasks. Global TFT remains active for 24 and 168 hourly targets with the original strong London cold-start evidence. The missing month capability is now served by a Chronos-2 120M model adapted through rank-8 LoRA. It improves the strongest frozen Tetouan selection baseline by 49.82%, reaches macro $R^2 = 0.4498$, and obtains 88.15% central-80% coverage. Portugal transfers positively, while the negative southern-Morocco diagnostic prevents an unsupported geographic claim. Applying the same architecture to 24 and 168 hours produces 2.90% and 2.62% gains over the packaged TFTs, but those challengers remain inactive until their extra context and memory costs are resolved.

The engineering contribution is EnergyAI, a Dockerized Next.js and FastAPI platform with PostgreSQL persistence, authenticated ownership, bounded CSV ingestion, 24/168-hour TFT and 30-day Chronos-2 capabilities, quantile output, history, analytics export, and readiness checks. The API and interface carry explicit target count, interval, resolution, input range, and model provenance. A new account can create 365 days of deterministic hourly simulation history, which satisfies every model lookback without weakening the 270-day gate or padding recent TFT context. The production image pins

and caches the exact foundation-model base, verifies checksums, and serves offline. The verified Azure jury release now serves all three model horizons; public readiness confirms that both TFTs and the daily Chronos-2 artifact are available, enabled, and warmed.

Several perspectives follow directly from the limitations. First, the models should be evaluated on Moroccan household or organizational data before local performance claims are made. Second, weather forecasts and static site attributes can be added through controlled ablations. Third, the Chronos runtime should use one shared base with switchable adapters before the hourly challengers are reconsidered. Fourth, finalists should be repeated over multiple deterministic seeds. Fifth, drift monitoring and governed retraining should be added without allowing unreviewed artifacts to replace packaged models. Finally, the public deployment should add a custom domain, security headers, managed identity, durable avatar storage, centralized monitoring, the remaining workers, backup-and-restore evidence, sustained provider and load testing, and disaster-recovery procedures.

The most important result of this work is not a percentage. It is an evidence-traceable transition from a task that revealed its answer to a system that states what it knows, what it predicts, how it was validated, and where its limits remain. That transition is the foundation on which future EnergyAI improvements should be built.

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Evidence and Requirement Traceability

A.1 Evidence policy

Every central claim in the report must resolve to a source file, manifest, CSV, notebook output, test log, or Git checkpoint. The machine-readable registries are stored under `report/evidence`. Their purpose is to prevent accidental promotion of an old presentation value, failed experiment, or unverified administrative detail.

Table A.1 – Report evidence registries.

Registry	Content
<code>model_registry.csv</code>	Model task, split, scaler scope, seed, artifact, metrics, status, and limitation
<code>artifact_hash_inventory.csv</code>	File sizes and SHA-256 digests for packaged and selected historical artifacts
<code>figure_registry.csv</code>	Figure source, generation method, evidence date, and status
<code>commit_timeline.csv</code>	Research and product checkpoints used in the narrative
<code>notebook_catalog.csv</code>	Notebook purpose, status, and methodological caveat
<code>test_evidence_summary.csv</code>	Current quality gates and limitations
<code>requirements_traceability.csv</code>	Requirement-to-code-to-test mapping
<code>lcl_tft_reanalysis_manifest.json</code>	Checkpoint/data/script hashes, reproduced cohort checks, and post-hoc quantile diagnostics
<code>lcl_tft_serving_normalization_manifest.json</code>	Preserved-data audit, exact serving-normalization comparison, hashes, environment, and generated evidence

Registry	Content
lcl _ tft _ serving _ normalization_comparison. csv	Research-versus-serving metrics for the frozen 24-hour and 168-hour cold-start cohorts
month _ production _ v3 / release/manifest.json	Frozen Chronos-2 base/adaptor identity, selection and transfer metrics, cadence, calibration, and run-time contract
hourly _ foundation _ v2 / protocol.json	Preregistered hourly challenger candidates, comparators, cohorts, and promotion gates
hourly _ foundation _ v2 / audit.json	Sealed cold-household accuracy, coverage, household-win, and FP32 reload results
hourly _ foundation _ v2 / operational _ benchmark . json	CPU load, memory, and warm-inference measurements used in the deployment decision
openapi _ inventory _ summary.json	Exact current OpenAPI path and operation counts
azure _ deployment _ evidence_2026-08-07.md	Release commit, Azure resource boundary, HTTPS endpoints, readiness checks, scale settings, cost controls, and remaining cloud limitations
azure _ deployment _ evidence_2026-08-09.md	Jury release commit, immutable image digests, email and Google capability, responsive smoke tests, retained rollback revisions, and remaining limitations
azure _ deployment _ evidence_2026-08-15.md	Three-horizon release commit and image digests, CI and remote build runs, Chronos-2 warm-up, Azure revisions, rollback handling, and public smoke results

A.2 Condensed requirement matrix

Table A.2 – Condensed requirement-to-evidence matrix.

Requirement	Implementation evidence	Validation evidence
Secure sign-in	Auth router, bcrypt service, access token, refresh-cookie flow	Authentication unit/integration tests and browser journey

Requirement	Implementation evidence	Validation evidence
Administrative RBAC	Admin router and role dependencies	Authorization tests
Per-user isolation	User-Site-Meter relationships and scoped queries	PostgreSQL integration suite
CSV import safety	Meter ownership, preview, bounded import, batch record	Ingestion tests and browser journey
Demo continuity	Deterministic 365-day hourly bootstrap, forward-only catch-up, source labels, and simulator-only reset	All-model input-readiness test and browser reset/status journey
Demo-to-CSV handoff	Fresh simulator-status check, explicit stop confirmation, preserved simulation history	Six-project browser matrix
Permanent account deletion	Reauthentication plus ordered set-based ownership-graph deletion	Full-history PostgreSQL deletion test and backend account-control tests
24-hour forecasting	Active Global TFT 24 h manifest and checkpoint	Artifact contract and dual-horizon tests
168-hour forecasting	Active Global TFT 168 h manifest and checkpoint	Artifact contract and dual-horizon tests
30-day forecasting	Active Chronos-2 LoRA month release with explicit daily cadence	Checksum, 365-day input, 30-target inference, API persistence, UI cadence, and PDF tests
Hourly challenger governance	Checksum-addressed Chronos-2 LoRA 24 h and 168 h releases	Frozen accuracy gates passed; resource/contract decision records why activation is deferred
Forecast provenance	Persisted Forecast entity and history endpoint	Forecast API tests and UI coverage
Alerts and actions	AlertConfig, Alert, Recommendation, workers	Alert state tests and worker health
Analytics PDF export	Forecast UI and analytics PDF endpoint; legacy reports route redirects to Usage	Horizon-specific PDF tests
Health/readiness	System live, ready, health, version endpoints	Docker health checks

Requirement	Implementation evidence	Validation evidence
Public deployment	Azure Container Apps, managed PostgreSQL, ACR, Key Vault, Google identity, and budget	HTTPS, readiness, model warm-up, active-revision health, email-worker health, OAuth origin handshake, and responsive browser smoke tests

A.3 Administrative information still required

Verified UMI/FSM identity, department, programme, and supervisor information have been incorporated. Still pending are the official Master PFE cover and formatting rules (the public faculty model is doctoral), jury composition, any required laboratory or host line, the defense date, and the approved AI-tool disclosure.

Research Assets and Checkpoints

B.1 Git chronology

Table B.1 – Key research and product checkpoints.

Date	Commit/tag	Meaning
2026-04-10	43f5cab	Project initialization
2026-04-12	ed3c48e	Initial paper text added
2026-04-15	b96fef0	First DEPM implementation
2026-05-02	8daf16c	Fair-comparison notebook and leakage investigation
2026-05-05	ea4e8de	Full six-experiment DEPM replication
2026-05-06	c40b120	No-leakage variant
2026-05-09	71ca537	Ten DEPM variants
2026-05-10	6f50535	Steel Industry cross-dataset experiment
2026-05-17	87a70c1	Continuous regression and critical findings
2026-05-17	a653dcf	CNN-BiLSTM continuous regression
2026-05-22	5999ecc	Forecasting application and baselines
2026-05-31	8813aca	PatchTST forecasting
2026-06-30	v0.1.0	Research checkpoint with notebooks and archived models
2026-06-30	c61cb66	Dynamic model registry
2026-07-08	phase2 forecast stable	- Stable registry forecasting -
2026-07-20	a6d1683	Truthful 24-hour TFT integration
2026-07-20	07b43b5	Fixed artifact-readiness contracts
2026-07-20	9525d3f	PFE release checkpoint
2026-07-22	9226b56	Gated 168-hour capability
2026-07-27	baacbf2	Both supported horizons shipped
2026-07-27	1e67921	Product V1 checkpoint

B.2 Notebook status

Table B.2 – Selected notebook status.

Notebook	Status	Interpretation
DEPM_Implementation.ipynb	Invalidated	Direct/proxy leakage and non-causal validation
DEPM _ Feature _ Ablation . ipynb	Diagnostic	Useful for identifying answer channels
LSTM _ Energy _ Predictor . ipynb	Non-comparable	Historical random-split classification
depm_final.ipynb	Invalidated	Derived/current target information remained
depm_final_noleak.ipynb	Partially corrected	Direct features removed; split/preprocessing caveats remained
fair_comparison.ipynb	Diagnostic	Train-only transform but classification task
proper _ regression _ forecasting.ipynb	Usable	Shifted features, chronological split, physical metrics
EECP _ CBL _ Replication . ipynb	Usable	Continuous chronological CNN-BiLSTM work
PatchTST _ Forecasting . ipynb	Usable	Direct causal forecasting
weather_sota_forecasting.ipynb	Incomplete	TSMixer success not proven by frozen metrics
ecl_deep_models_dual_t4_benchmark.ipynb	Partially complete	Four model families have completed rows; Time-Pro/TimeMixer++ do not
lcl - global - household - forecasting - full - selected-pro.ipynb	Complete	Source for final household candidates and production selection

B.3 Checkpoint classification

— **Historical invalidated:** old DEPM DNN, ResNet, PCA, scaler, XGBoost, and classification LSTM assets.

- **Research archive:** PatchTST, advanced PatchTST, iTransformer, CNN-BiLSTM, and multi-horizon checkpoints available at historical Git tags.
- **Benchmark evidence:** ECL and Low Carbon London leaderboards and their run metadata.
- **Candidates:** Global N-BEATS and MultiCycleNet-adapted experimental outputs.
- **Packaged active checkpoints:** `global_tft_24h`, `global_tft_168h`, and `month_production_v3`; frozen research, transfer, and serving-normalization metrics are reported under distinct protocols.
- **Packaged inactive challengers:** `hourly_foundation_v2/release/24h` and `168h`; both pass their frozen London gate, but deployment is deferred under the recorded resource and input-contract decision.

B.4 Production hyperparameters and contracts

Table B.3 – Active artifact contract summary.

Property	24-hour artifact		168-hour artifact	
Lookback	336 h		336 h	
Horizon	24 h		168 h	
Quantiles	0.1, 0.5, 0.9		0.1, 0.5, 0.9	
Trainable parameters	1,386,262		1,386,262	
Checkpoint bytes	5,600,105		5,600,105	
Minimum coverage	95%		95%	
Maximum consecutive gap	3 h		3 h	
Research-evaluation normalization	Per-household segment	training-z-score	Per-household segment	training-z-score
Serving normalization	Latest-window score	336-hour z-score	Latest-window score	336-hour z-score

The weekly manifest records hidden size 128, four attention heads, dropout 0.1, and 186 state-dict tensors. The original worker and strict checkpoint load confirm the same architecture for the two frozen runs. Metric equivalence was not assumed from file compatibility: the rolling serving normalization was evaluated separately on the exact preserved cold-start samples.

B.5 Checkpoint reanalysis provenance

Table B.4 – Full hashes and sources for the post-hoc TFT diagnostics.

Item	Repository path	SHA-256
24-hour checkpoint	models/lcl_global_forecasting/full_selected_v1/runs/global_tft/day_24h/best.pt	60fedcdee375dc0b2973e55b9f4ec752da69c2a7e390e1f951ed5cbb7bc5a04d
168-hour checkpoint	models/lcl_global_forecasting/full_selected_v1/runs/global_tft/week_168h/best.pt	80af16b25af9b912019c6e40cfdc491e2cf5173e5743144596801e28df695f93
Hourly values	models/lcl_global_forecasting/full_selected_v1/data/hourly_raw.npy	353d5e21355ada4dd662b74474f021afabcb2300cd1f3cf934a3792bc6487646
Hourly calendar	models/lcl_global_forecasting/full_selected_v1/data/hourly_calendar.npy	32ac697bf3890b924572fca92b029b9b64cce9957f693d5c3aa985fc375a4b41
Hourly scalars	models/lcl_global_forecasting/full_selected_v1/data/hourly_scalars.npz	401e2b889fd7f94fe4091c891e4a20c0ce1447c730f81d0be5a71eb53e88a16a
Reanalysis script	report/source/analyze_lcl_tft.py	4bce1b95797f82fe2672e3b82d607599a0eca1ba06d99147564dd9129ac67cb4
Prepared metadata	models / lcl _ global _ forecasting / full _ selected _ v1 / data / prepared.json	b7bec0bc69af3d9af6c5a9855fc4f6055b737fee31cc70e54ac98da2cd27aa1f
Cold assignments	models/lcl_global_forecasting/full_selected_v1/data/cold_households.npy	4edd0dbc7b9d89944a0649edaf21535585a276b81c086cec73248bd1dd015444
Household identities	models / lcl _ global _ forecasting / full _ selected _ v1 / data / households.json	5e83b3ab8dc62680e8cc51e63530cb96a4b7eada8909c89d0dc154cb6fbd7519
Serving-evaluation script	report/source/evaluate_lcl_serving_normalization.py	ffdd74c33a7403e6f3636104efbaf44c558dea6cd5de6fc2204af39e8f81f7f7
Production preprocessing source	backend/app/services/product_forecast_service.py	312baaf589c5314cdfb747c99002a6203d2f6d9358f46476a12cb8aa1147dae3

The diagnostic manifest was generated on 1 August 2026 at repository commit `1e67921624c7dd7bd1b1ef9b93fc00ddcb28e09f`. It records 11,871 windows from 500 households for 24 hours and 11,830 windows from 499 households for 168 hours. The script validates its recomputed macro MAE and exact household win counts against the frozen metrics before writing any report evidence.

The serving-normalization manifest was generated on 1 August 2026 at repository commit `f6679d4b5f759511dcd8bec26745bb894633f66a`, with an explicitly dirty working tree because the evidence script and report revision were not committed. It records the exact fixed origins, prepared-data hashes, checkpoint equality, execution environment, production preprocessing function, CPU-versus-GPU parity checks, per-household CSVs, and the research-versus-serving comparison. No experiment or production artifact was overwritten.

Technical Contracts, UML, Tests, and Installation

C.1 Principal API contracts

Table C.1 – Selected API operations.

Endpoint	Method	Purpose
/api/v1/auth/login	POST	Authenticate and create access/refresh session
/api/v1/auth/refresh	POST	Rotate refresh state and return a new access token
/api/v1/settings/setup	POST	Create the initial site/meter configuration
/api / v1 / ingestion / meters / {id} / csv / preview	POST	Validate a bounded CSV preview
/api / v1 / ingestion / meters/{id}/csv/import	POST	Confirm and persist accepted readings
/api / v1 / simulation / status	GET	Report simulator continuity and labelled history state
/api / v1 / simulation / start	POST	Bootstrap if empty, catch up if eligible, and start live simulation
/api / v1 / simulation / reset	POST	Replace simulator-owned history without deleting CSV or push rows
/api / v1 / consumption / period	GET	Summarize consumption for a selected period
/api / v1 / forecast / capabilities	GET	List supported and enabled horizons
/api / v1 / forecast / readiness	GET	Explain artifact/history readiness

Endpoint	Method	Purpose
<code>/api/v1/forecast/run</code>	POST	Generate and persist a supported forecast
<code>/api / v1 / forecast / history</code>	GET	Return owned forecast history
<code>/api/v1/alerts</code>	GET	Return owned alert records
<code>/api / v1 / analytics / report/pdf</code>	GET	Generate the supported analytics PDF
<code>/api/v1/account/delete</code>	DELETE	Reauthenticate and permanently remove the owned account graph
<code>/api/v1/system/live</code>	GET	Process liveness
<code>/api/v1/system/ready</code>	GET	Dependency readiness

The complete current OpenAPI schema contains 80 method–path operations across 74 unique paths. This count was extracted from `backend/app/main.py` at commit `1e67921`. The full commit and extraction method are stored in `report/evidence/openapi_inventory_summary.json`.

C.2 UML diagrams

This report deliberately includes only use-case, class, and sequence UML types. The main chapters contain the use-case, core domain class, authentication sequence, CSV import sequence, and forecast sequence diagrams. They are generated programmatically from a text-source Matplotlib script and registered with their repository sources.

C.3 Test plan

Table C.2 – Condensed test plan.

Layer	Technique	Current evidence
Schemas/services	Unit and focused integration tests	Validation, auth, settings, forecast, alert, report, and account behavior
Database/API	Full pytest suite on PostgreSQL	100 passed, 0 skipped
Database/API	Current local pytest suite	116 passed, 4 PostgreSQL-only skips

Layer	Technique			Current evidence
Database/API	Current	CI	Post-	Passed
	greSQL and test-image gates			
Frontend static	ESLint and TypeScript			Both passed
Frontend build	Next.js	production		Passed, 25 routes
	compilation			
Browser E2E	Playwright	six-project		198 passed, 0 failed
	matrix			
Model artifacts	Hash,	strict load,	infer-	Active 24 h, 168 h, and 30-day re-
	ence shape,	finite val-		leases validated; hourly challengers
	ues, cadence			audited separately
Dependency secu-	npm and	production		Zero known vulnerabilities
rity	pip audits			
Deployment	Docker	health	plus	Public web/API, email worker,
	Azure HTTPS/browser			managed database, both TFTs, 30-
	smoke			day Chronos-2, SMTP capability,
				and Google origin verified

C.4 Docker installation guide

C.4.1 Prerequisites

Install Docker Desktop with Compose support and Git. Obtain the repository and create an environment file from the supplied example. Replace all sentinel values for database password, JWT secret, administrator password, and provider settings. Do not reuse demonstration secrets in a public environment.

C.4.2 Startup

```
git clone https://github.com/BR1WA/energy-forecast-platform.git PFE2
cd PFE2
copy .env.example .env
# Edit .env and replace every insecure sentinel value.
docker compose up --build -d
docker compose ps
```

Verify:

```
curl http://localhost:8000/api/v1/system/live
curl http://localhost:8000/api/v1/system/ready
curl http://localhost:3000/
```

The verified Azure deployment supplies managed TLS, Azure hostnames, private images, externalized secrets, a healthy email worker, ready SMTP capability, and a production Google identity origin. Web, API, and email worker use revision suffix `v4-d25295b`. The backend image caches the pinned foundation base, enables `FORECAST_30D_ENABLED`, and passed model warm-up at the API's two-vCPU and 4-GiB allocation. Public readiness reported the 24-hour, 168-hour, and 720-hour artifacts available, enabled, and warmed. Remaining production work includes an application security-header policy, custom domain, managed identity, durable avatar storage, centralized monitoring, Azure alert/avatar workers, and a tested backup-restore procedure.

C.5 Report reproduction

The report sources are modular. From `report/source`, run:

```
python generate_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The final build process also runs Biber, renders the PDF with Poppler, checks log warnings, searches for unapproved placeholders and local paths, and packages the source tree.